

The Straits of
Malacca *and* Singapore

— A Model for the World —



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A Model for the World

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Foreword

o/s

I welcome this book. It is edited by four eminent experts on the law of the sea and the contributors are very knowledgeable on the topics assigned to them.

My good friend, Professor Tommy Koh, and I spent several years of our lives as members of Singapore delegation to the UN Seabed Committee and later to the Third UN Conference on the Law of the Sea (UNCLOS). One critical issue was how to ensure that the then prevailing unimpeded navigational freedoms in straits used for international navigation (SUIN), could be preserved and safeguarded in the context of coastal states' extension of the limits of their territorial waters to 12 nautical miles.

Singapore worked closely with other like-minded delegations to get UNCLOS to enshrine the concept of "transit passage" into the new Convention.

We live in troubled times. The crisis in the Strait of Hormuz has highlighted the issue of freedom of navigation in SUIN.

This book, therefore, is very timely.

The Straits of Malacca and Singapore is one of the most important SUIN, with strategic importance and as well as importance for shipping, trade and commerce. At its narrowest point, it is barely 3.5 kilometres wide. Yet, through this narrow SUIN passes a substantial share of global seaborne trade and energy through container vessels and crude carriers. It is one of the most heavily trafficked sea passages in the world.

The book examines the history, international legal regime, hydrography, security, marine environment, commercial relevance, co-operation between littoral states and measures adopted through the International Maritime Organization of the Straits of Malacca and Singapore. Each of these disciplines is often viewed and discussed in isolation from others, yet the editors have successfully resisted against siloed examination, making this book a strong and comprehensive account of the Straits of Malacca and Singapore.

Written in an approachable manner, it bridges specialised interest of lawyers, diplomats, policymakers, and mariners, and general interest of any member of the public.

It serves as a reminder to the world of the importance of major global maritime chokepoints in the flow of global energy and trade. While the Straits of Malacca and Singapore may not face identical geopolitical tensions as Hormuz, comparison reveals perpetual questions and issues that major maritime chokepoints face, such as ensuring that passage remains unimpeded, the importance of co-operative arrangements to mitigate the vulnerability of passage, and the consequences should such major maritime chokepoints face closure.

Professor S. Jayakumar

July 2026

About the Editors

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He had served as Dean of the Faculty of Law of NUS, Singapore's Permanent Representative to the United Nations (UN) in New York, Ambassador to the United States of America, High Commissioner to Canada and Ambassador to Mexico. He was President of the Third UN Conference on the Law of the Sea. He was also the Chairman of the Preparatory Committee for and the Main Committee of the UN Conference on Environment and Development (Earth Summit). He was the UN Secretary-General's Special Envoy to Russia, Estonia, Latvia and Lithuania.

Nilüfer ORAL is Director of the Centre of International Law (CIL) at the National University of Singapore (NUS). She is a member of the United Nations International Law Commission (ILC) and served as Co-chairperson of the 74th session of the Commission. She was also a Co-chair of the ILC Study Group on sea-level rise in relation to international law. In addition, she is a member of The Hague Academy of International Law Curatorium and an associate member of the Institut de Droit International. She was a legal advisor and climate change negotiator for Turkish MFA (2009–2016). She is currently a Member of the Committee of Legal Experts of the Commission of Small Island States on Climate Change and International Law (COSIS). She is a Distinguished Fellow of the Law of the Sea Institute at Berkeley Law and a Senior Fellow of the National University of Singapore Law School. She is currently a member of the Steering Committee of the World Commission on Environmental Law, and also a member of the Board of Governors of the International Council of Environmental Law. She has appeared before the International Tribunal for the Law of the Sea (ITLOS) and the International Court of Justice (ICJ) for advisory proceedings and contentious cases. Dr Oral is the series co-editor for the *International Straits of the*

World publications (Brill) and on the editorial board of different publications.

Robert BECKMAN is an Emeritus Professor at the Faculty of Law at the National University of Singapore (NUS), where he has taught courses in public international law since 1977. He was the founding Director of the Centre for International Law (CIL), a university-level research institute at NUS that was established in 2009, and he is currently a Senior Advisor to its Ocean Law and Policy Programme. He is also an Adjunct Senior Fellow and Adviser with the Maritime Security Programme of the Institute for Defence & Strategic Studies (IDSS) at the S. Rajaratnam School of International Studies (RSIS) at Nanyang Technological University (NTU).

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International Law and Ocean Development and International Law Journal. She received a Fulbright Scholarship in 2013 and the NUS Overseas Graduate Scholarship in 2014. She is a qualified Advocate and Solicitor in Singapore. Her research interests are in public international law and law of the sea, and she has written extensively on a variety of issues, including on the international regulatory regime governing submarine cables and pipelines.

About the Contributors



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Dr **Peter BORSCHBERG**, FRHistS, FRGS, is an independent researcher based in Singapore. He has published extensively on early modern trade, diplomacy, and Euro-Asian interaction, with a particular focus on the Singapore Straits and the wider Malay-Indonesian world. His research explores maritime history, the law

of nations, and the political and commercial networks that shaped Asia's global connections in the 15th to 18th centuries.

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Dr **Parry OEI** was Director of the Port Services Division and former Chief Hydrographer in the Maritime and Port Authority of Singapore.

He held the following key positions in regional and international organisations:

- Chair of the International Hydrographic Organization's Inter-Regional Coordinating Committee;
- Chair of the East Asia Hydrographic Commission; and
- Chair of the Tripartite Technical Experts Group's Marine Electronic Highway Working Group in the Straits of Malacca and Singapore.

He is currently the Co-Chair of the United Nations Global Geospatial Information Management and International Hydrographic Organisation Joint Working Group on Marine Geospatial Information.

Ambassador **Mary SEET-CHENG** was a diplomat in the Ministry of Foreign Affairs of Singapore (MFA) from 1973 to 1997 and served in the Singapore Missions in Canberra, London and Brussels. She was Ambassador to the Benelux countries, the European Commission, and the Holy See from 1993 to 1996. In 1997, she was seconded to the Maritime and Port Authority (MPA) of Singapore as Director (Policy). She remains MPA's Special Adviser. In 2006, she returned to MFA as Senior Specialist Adviser (SSA) on maritime matters and was concurrently Singapore's non-resident Ambassador to the Republic of Panama and Cuba from 2007 to 2020. She has been Singapore's Non-Resident High Commissioner to the Republic of Fiji and Non-Resident Ambassador to the Pacific Islands Forum since 2020.

Chapter 1

An Introduction to the Evolution of the Law of the Sea on Straits Used for International Navigation

o/s

Nilüfer Oral

Introduction

Why do straits matter for the law of the sea? Why do they matter to the world, and why do the Straits of Malacca and Singapore (SOMS), the topic of this book, matter in particular?

First, straits, especially the Straits of Malacca and Singapore, have played a formative role for the law of the sea starting in the year 1603 with the attack and capture by the Dutch of the *Santa Catarina*, a Portuguese ship, in the Straits of Malacca and Singapore. Today, this

is where Changi Airport is located. This event launched the career of a young Dutch lawyer, called Hugo Grotius, who later acquired fame as the defender of the freedom of the seas in his classic treatise “Mare Liberum” (the Free Sea) in 1609. Freedom of the sea is one pillar of the law of the sea.

In more modern times, the law of the sea was influenced by another strait – the Corfu Channel, which is found between Albania and Greece. It was the subject of the very first case brought before the newly established International Court of Justice. The case was filed by the United Kingdom against Albania in 1947 and stands as a landmark case affirming the right of nonsuspendable innocent passage for all foreign flagged ships, including foreign warships, in straits used for international navigation.

Second, the world economy depends on the free movement of goods across thousands of miles. Straits have been an essential part of world trade for centuries. They are the maritime arteries for international shipping and global commerce. Some 80 per cent of global trade relies on shipping, which is far more cost-effective than transport by air or land. Container ships can carry tons of cargo at a fraction of the cost of airplanes. Trains are

landlocked and cannot cross continents. Straits used for international navigation significantly reduce the time and cost of sea travel connecting seas because they create narrow maritime short-cuts linking seas. A good example of the importance of international straits for global trade are the Straits of Malacca and Singapore, which carries 25 per cent of global trade, including oil and gas. This is why there has been much interest in the melting Arctic, as the opening of year-round navigation through the Arctic straits would reduce the distance and time for shipping significantly and save millions of dollars.

Third, world energy supplies depend on straits. There are several straits that are considered “vital chokepoints” for energy transport, i.e., oil and gas, including the Straits of Malacca and Singapore. For example, the Strait of Hormuz is a critical artery for the transport of 20 to 30 per cent of oil and gas, much of which is Asia-bound. The Iran-US-Israel conflict has made the world experience first-hand the global consequences of closing a major strait used for international shipping. The Straits of Malacca and Singapore are considered the most important chokepoint for the global transport of oil and gas.

The Straits of Malacca and Singapore

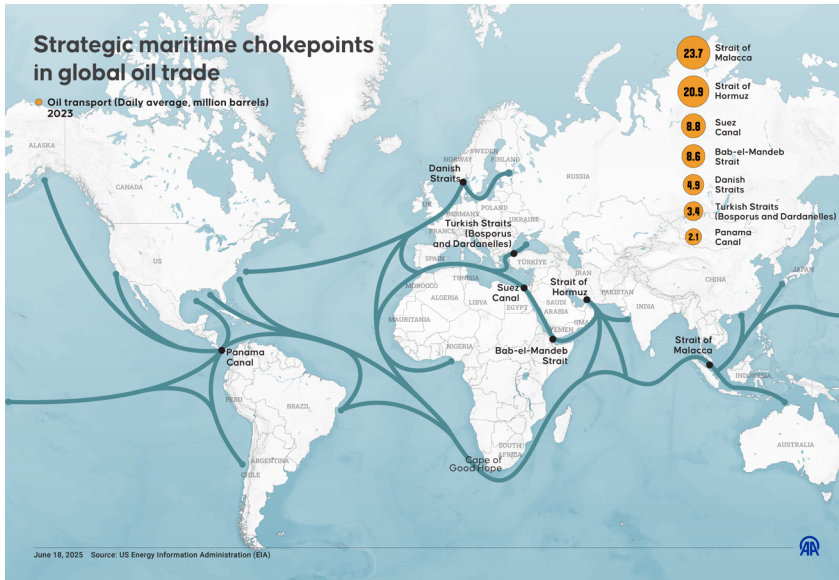


Figure 1.1: Strategic maritime chokepoints in global oil trade.

Source: Anadolu Ajansı.

Fourth, straits also are critical for food security. This became most evident when Russia invaded Ukraine, where bombings prevented the shipping of much-needed global grain supplies from the port of Odessa in the Black Sea and through the Turkish Straits. Some 55 per cent of internationally traded maize, wheat, rice and soybean is shipped through at least one maritime chokepoint. Eleven per cent relies on the Turkish Straits and the Strait of Hormuz, for which no alternative route exists. Annual throughput of grain ranges from 24 million tonnes (via the Strait of Hormuz) to 108

million tonnes (via the Straits of Malacca and Singapore). In 2000, 42 per cent of global grain exports was shipped through one or several of the maritime chokepoints. In 2015, that total had risen to 55 per cent.

Straits and the Modern Law of the Sea

The legal regime of straits used for international navigation has always been closely tied to the question of the breadth of the territorial sea. For this reason, the regime of straits can be divided into two distinct periods of time: before the adoption of the 1982 United Nations Convention on the Law of the Sea (UNCLOS) and after. Before UNCLOS, the regime for straits was binary: non-suspendable innocent passage for straits that were part of the territorial sea of a coastal State and high seas freedom of navigation for that part of straits that came within the high seas. Until UNCLOS, States could not come to an agreement on the breadth of the territorial sea. The 1958 Geneva Convention on the Territorial Sea and Contiguous Zone did not specify a maximum breadth. It recognised non-suspendable innocent passage for straits that were part of the territorial sea of the coastal State. Another attempt was made in 1960 where a proposal for a six-nautical-mile territorial sea

failed by one vote. Then came UNCLOS, which was a game changer for the law of the sea, and especially for straits used for international navigation.

Before UNCLOS was adopted, the oceanscape was changing as coastal States jockeyed for greater control over larger ocean space, creating disputes between States. The dispute over the breadth of the territorial sea was not resolved in the First United Nations Conference on the Law of the Sea or the Second. The Third United Nations Conference on the Law of the Sea (UNCLOS III) was launched following the historic speech by Ambassador Arvid Pardo from Malta. The negotiations lasted from 1973 to 1982. Ambassador Tommy Koh of Singapore became the President of UNCLOS III following the untimely passing of Hamilton Shirley Amerasinghe of Sri Lanka in 1980. The result of nine years of negotiations produced a very innovative and landmark convention, which Tommy Koh famously called the “Constitution of the Ocean”. Among the most important innovations introduced by UNCLOS was Part III on the regime of straits used for international law.

The question of straits was one of the more contentious issues during the negotiations. A proposal for a

12-nautical-mile territorial sea was contingent upon resolving the knotty issue of whether innocent passage or high seas freedom of navigation would continue to apply in straits that would be enveloped by a 12-nautical-mile territorial sea. Expanding the territorial sea to 12 nautical miles would enclose a vast area of sea that was high seas and subject to the freedom of navigation regime. Indeed, some 100 straits would be affected. This set off a key debate in UNCLOS III, which could be called “the Great Straits Debate”.

It is no surprise that a 12-nautical-mile territorial sea proposal triggered opposition from the major maritime powers, such as the United States, who sought to protect freedom of navigation, in particular for aircraft and warships, including submarine passage. In the high seas, there are no restrictions on the passage of warships and aircraft. Whereas, according to the innocent passage regime, foreign submarines must engage in surface passage. In addition, there is no right of overflight for aircraft under the innocent passage regime.

On the other side was a group of strait States who advocated for maintaining the innocent passage that applied to the territorial sea. These States were concerned

that the increase in shipping, in particular oil tankers, would result in more pollution. There were also concerns about nuclear waste shipments and nuclear-powered submarines. These states wanted to regulate safety of navigation and protection of the marine environment. The division threatened to block progress until skilful diplomacy stepped in.

Ambassador Satya Nandan of Fiji is recognised as one of the architects of what became an entirely new “transit” passage regime. Together with the United Kingdom, a new transit passage for straits used for international navigation was proposed as a compromise that would balance the two competing positions. This was known as the UK/Fiji proposal. It is found in article 37 of UNCLOS and applies to straits that connect one part of the high seas or the exclusive economic zone (EEZ) of a coastal State with another part of the high seas or EEZ. Examples of such straits are the Straits of Malacca and Singapore, the Dover Strait, the Strait of Gibraltar, and more.

However, the compromise also excluded several straits from the transit passage regime. These are straits where there is an alternative route for ships of similar convenience, or straits that only connect to a territorial

sea or another State. In addition, straits used for international navigation that are subject in whole or in part to a long-standing international convention are entirely excluded from Part III, such as the 1936 Montreux Convention on the Regime of the Straits for the Turkish Straits. Finally, UNCLOS for the first time allowed archipelagic states to enclose an area of ocean space that fulfilled a set of requirements as archipelagic waters, similar to the territorial sea of a coastal State. This was subject to archipelagic sea-lane passage that is akin to the new transit passage adopted under UNCLOS.

Transit Passage versus Non-Suspendable Innocent Passage

What are the differences between transit passage and non-suspendable innocent passage? Innocent passage is defined as passage that “is not prejudicial to the peace, good order or security of the coastal State.” Transit passage is simply defined as “the exercise in accordance with this Part of the freedom of navigation and overflight solely for the purpose of continuous and expeditious transit of the strait between one part of the high seas or an exclusive economic zone and another part of the high seas or an exclusive economic zone.”

The Convention enumerates many specific acts that are deemed non-innocent passage, allowing the coastal State to interfere with the passage of the ship. There is no similar corollary list of acts that would deem a violation of transit passage other than saying “Any activity which is not an exercise of the right of transit passage through a strait remains subject to the other applicable provisions of this Convention.” An important concession to the maritime powers was the preservation of the freedom of overflight and submerged passage for foreign submarines. Foreign submarines in innocent passage are required to engage in surface passage and show their flag. There is no restriction on submarine passage for transit passage.

The compromise reached in the new transit passage balanced the broader navigational interests for foreign ships by allowing some regulatory competence to the strait State. Under the innocent passage regime, the coastal State adopt laws and regulations for the safety of navigation, the regulation of maritime traffic, the protection of navigational aids and facilities, and other facilities or installations. In addition, the coastal State can adopt laws and regulations to protect cables and pipelines. There is also a separate provision that allows

the coastal State to regulate the passage of foreign nuclear-powered ships, and ships carrying nuclear or other inherently dangerous or noxious substances shall, when exercising the right of innocent passage through the territorial sea, carry documents and observe special precautionary measures established for such ships by international agreements.

By contrast, the transit passage only provides for the competence of the strait State for the safety of navigation and the regulation of maritime traffic, but only to establish traffic separation schemes or sea lanes with the approval of the International Maritime Organization. However, such approval is not required for traffic separation schemes or sea lanes that are established in territorial sea and subject to innocent passage.

Another important difference is the narrower competence to regulate protection of the marine environment in the case of transit passage. While innocent passage gives the coastal State broad competence to adopt laws and regulations to *preserve the environment of the coastal State and for the prevention, reduction, and control of pollution, including the conservation of marine living resources of the sea*, transit passage is limited to adopting laws and

regulations that give effect to “applicable international regulations regarding the discharge of oil, oily wastes and other noxious substances in the strait”. This would restrict the strait State to only implementing the International Convention for the Prevention of Pollution from Ships (MARPOL). It also excludes conservation measures for marine living resources. But in fairness, during the UNCLOS III years, vessel-source pollution and notably those related to oily wastes and oil pollution were the predominant concerns.

Conclusion

Straits used for international navigation provide cost-effective shipping routes and are critical for the global economy. Until the 1982 UNCLOS, the legal regime of passage was either non-suspendable innocent passage or high seas passage. That changed when a proposal for a 12-nautical-mile breadth of the territorial sea was made during UNCLOS III. Through skilful negotiations, creative thinking, and compromise, a new transit passage regime was created to balance the competing interests of maritime powers and strait States.

Chapter 2

The Singapore and Melaka Straits in Historical Perspective

✎

Peter Borschberg

The Straits of Malacca and Singapore form natural boundaries. They sit between Singapore, Malaysia, Thailand, and Indonesia as if their function were self-evident: thin strokes of ink separating territorial blocks. The eye reads these maps effortlessly. North is above, south below. Sovereignty is contained within shaded land spaces. The sea is rendered as blue space: the gap between states.

The border function of the Straits is a comparatively recent development and dates to the Anglo-Dutch Treaty of 17 March 1824. This treaty settled outstanding issues between the British and the Dutch and established

spheres of influence for them: the north was allocated to the British and the south to the Dutch. The Singapore and Melaka Straits became the notional dividing line that separated the two imperial powers. At first glance, the line of division might appear plausible and perhaps even natural, but from a historical vantage point, it marks a radical intervention. The line of division converted a busy maritime artery into a border zone.

It is easy to miss the scale of this transformation as we instinctively describe the Straits as a “frontier”. The word feels neutral. It is not. The frontier concept carries a specific historical inheritance, most famously associated with Frederick Jackson Turner. In that tradition, the frontier advances settlement through a moving line where settlers expand outward into a space imagined as less organised. It is land-based, directional, and teleological. Even when the idea of a frontier is criticised, it often retains its basic architecture: it is a line, it separates two zones, it moves in one direction, and it culminates in consolidation.

When this concept is imported into the imperial histories of Asia, the frontier is perceived as the outer rim of metropolitan authority, in which a powerful centre expands and takes over new land. The story thus

becomes one of expansion and stabilisation. It is misleading when applied to the Singapore and Melaka Straits. The Straits appear historically as the frontier between Siam and the Malay world, the frontier of Javanese influence, the frontier of Portuguese ambition, the frontier of Dutch East Indian consolidation, the frontier of the British Empire. Instead of being seen as important in itself, the sea is treated as a marginal space that awaits definition.

For the period before the 19th century, such a framing is deceptive. The idea of a frontier assumes that power radiates outwards from a territorial centre. Yet in the western archipelago, power was rarely territorial. Authority did not primarily depend on surveyed boundaries or fixed jurisdictions. It operated through webs of relationships. It was inherently relational, where allegiances could be fluid and shift. Tributary ties could overlap. Naval capacity, commercial leverage, and diplomatic recognition often mattered more than any projection of authority.

In that world, the Singapore and Melaka Straits were not empty edges awaiting incorporation. They were densely used maritime corridors that have evolved centuries before European boundary-making. Ships

moved through them seasonally and predictably. Ports functioned as hubs facilitating trade. Maritime communities shaped the flow of goods and information. To call such a space a “frontier” is to impose a 19th century territorial logic onto a 15th or 16th century maritime world, which regards the maritime space as one that is only meant to be divided, ignoring the maritime space for what it was.

To do justice to the pre-19th century history of the Straits, we need to drop the frontier and adopt a maritime-centric perspective. Instead of beginning with land-based polities and treating water as a separator, the primary starting focus is the sea as an organising medium. We should ask: What made these waters navigable, valuable, and governable? The answer can be found in the lack of viable maritime alternatives (other than the Sunda and Bali Straits), monsoon rhythms, currents, anchorages, shoals, fresh water, and the positioning of islands. This moves away from the idea of territory to the pragmatism and feasibility of maritime routes. The Straits then come into view not as boundary lines but as connective tissue.

In their earlier history, therefore, the Singapore and Melaka Straits did not divide, they connected. They

functioned as corridors of movement, circulation, and exchange. Today, as in the past, they link the Indian Ocean with the Pacific, and the Bay of Bengal with the South China Sea. Long before steamships and container vessels, the monsoon winds carried traders, pilgrims, envoys, and adventurers through the same maritime arteries. The seasonal reversal of winds did not simply shape navigation. It organised economic and diplomatic timetables. It produced predictable seasons for arrival and departure.

In this configuration, the Straits were certainly not marginal edges of nations but functional hubs where regional and trans-regional trade overlapped. They formed a hinge between South and East Asia, between Islamic commercial networks of the western Indian Ocean and the mercantile circuits of southern China. Gujarati merchants, Tamil traders, Arab navigators, Chinese junks, Javanese vessels, and later, Portuguese galleons moved through the very same corridor. These trajectories converged in harbours and marketplaces that accommodated multiple peoples, languages, and commercial practices. Pre-colonial Malay texts, West Asian travel accounts, Chinese records, and later European reports broadly all converge in recognising

this: the Straits were not empty water between meaningful lands. They were highly meaningful in themselves.

From the deck of a vessel, geography appears differently ordered. Land recedes. Coastlines become navigational guides rather than political edges. What matters are currents and counter-currents, the depth of anchorages, the location of shoals and sand banks, the availability of food and fresh water, and the protective contours of islands. In this maritime world, at different times, ports such as Melaka, Kedah, Johor, Aceh, Bintan, and Singapore were not seen as isolated centres but formed part of a bigger picture: interfaces where inland resources met maritime circulation, where goods were transhipped, information exchanged, alliances negotiated and authority performed. The Straits stitched together disparate shores into a shared economic and political fabric. Only later would they be redrawn as lines of separation.

A maritime-centric perspective does more than redirect our attention. It fundamentally changes the questions we ask. This chapter, therefore, aims to frame the historic functionality of the Singapore and Melaka Straits through two distinct but interrelated concepts: the contact zone and contested space. They highlight

different dimensions of the same maritime reality. Together they allow us to recover the Straits as a structured region of circulation, interaction, and rivalry rather than as a frontier.

The idea of a *contact zone* helps us to ask how interaction unfolded. It directs attention to sustained encounters among cultures, languages, legal systems, political authorities, and economic actors, often under conditions of asymmetry. In this framework, the sea is not a corridor between pre-existing worlds. It is a setting in which worlds meet and are reshaped through diverse and repeated forms of contact: through translation and misunderstanding, co-operation and conflict, adaptation and innovation. The concept gained methodological traction when maritime historians began to treat shared sea basins as historically structured spaces rather than as watery expanses separating states. The point is simple but decisive: Maritime contact zones often serve as areas where ships are not just carriers of goods and people, but also mobile platforms of exchange and diplomacy. Order arises not so much from imposing authority, but through the regular flow of goods and peoples.

Applied to the Singapore and Melaka Straits, the contact zone model shifts emphasis from boundary to interaction.

The Straits linked the Bay of Bengal, the Java Sea, and the South China Sea as subparts of a single maritime continuum. Vessels arriving from the Coromandel Coast of India carried textiles and bullion. Ships connected to Red Sea networks carried goods and religious linkages with Mecca. Chinese junks brought ceramics, silk, and sometimes diplomatic missions. Sailing craft from Java and the eastern archipelago moved foodstuffs, forest products, and people. Over centuries, these different circuits of maritime trade converged in the Singapore and Melaka Straits, resulting in encounters, exchanges, and occasionally in collisions.

What emerged was a plethora of rules and practices. Islamic commercial law intersected with adat and local customary practice. Chinese mercantile conventions shaped credit and partnership. Later, European legal doctrines and licensing regimes attempted to organise this without fully displacing existing norms. Merchants navigated overlapping jurisdictions as practically as they navigated winds and shoals. Political practice was similarly layered. Tribute missions co-existed with private trade. A ruler might receive an embassy and perform hierarchy while merchants pursued profit in the same harbour. The Straits thus engendered proximity

among the different actors and required sustained mediation: interpreters, brokers, pilots, patrons, and intermediaries. In this configuration, the Straits are best understood not as margins of pre-modern polities, but as critical nodal points within a wider maritime matrix.

Contact, however, does not tell the whole story. Shared spaces do not only connect. They also generate rivalry. The contested space lens, therefore, asks not only who meets here, but also who sets claim to the space. It foregrounds competition and negotiation over authority. A maritime corridor that concentrates wealth and mobility becomes a field in which influence is asserted, contested, and recalibrated.

To describe the Singapore and Melaka Straits as *contested spaces* is to recognise that control over these waters was never settled. Authority was personal, layered, and overlapping. Several polities advanced claims that were performed, resisted or negotiated over time. In the western Malay Archipelago before the 19th century, power was not primarily territorial in the modern sense of understanding. It was deeply relational. Authority depended less on enclosing land behind rigid borders than on managing relationships across and especially

above the water. It rested on the capacity to attract followers and merchants, mobilise naval forces, regulate trade, earn recognition, and project prestige.

In such a system, authority was elastic. It expanded and contracted with political fortunes, naval strength, and commercial reputation. It could be strong along particular routes but weak elsewhere. It was enacted through tribute, alliance, ritual hierarchy, and maritime capacity rather than through territorial demarcation. The Straits were contested not because they lay at the edge of empires, but because they were vital to the passage of ships and trade. Whoever influenced the flow of ships influenced the flow of wealth. Whoever provided protection commanded loyalty.

This logic shaped behaviour across the region. Malay sultanates competed to attract merchants by offering security and predictable regulation, because commercial confidence meant credibility and influence. Siam projected tributary claims southward through demands for recognition and ritual subordination rather than through occupation. Javanese polities pursued influence across island networks through alliance, kinship, and control over maritime manpower. European powers entered this already contested field and attempted to

redirect it. The Portuguese, after seizing Melaka in 1511, sought to regulate movement and circulation of goods and peoples through forts and licensing systems. The Dutch East India Company pursued monopoly arrangements within existing circuits, relying on treaties, patrols, and local allies. The British subsequently sought a different approach through clear demarcation of maritime territory for stability. Yet even this shift was not the natural end point of a frontier process. It was re-ordering: the replacement of negotiated maritime authority with demarcated separation and confirmed by maps, charts and measured co-ordinates.

Taken as separate but related concepts, the contact zone and contested space show what these maritime spaces really were. The Straits were arenas where order was produced through interaction and rivalry, where authority was enacted through mobility and protection, and where authority was negotiated rather than mapped. They also clarify what was so radically new about the arrangements of 1824. The pencil line drawn on the map of Asia by the chief Dutch negotiator, Anton Reinhard Falck, did not merely settle lingering disputes. It transformed a historically connective corridor into a boundary, translating a maritime world of routes and relations into a territorial world of demarcation.

Recovering the earlier emphasis on maritime space allows the Straits to reappear in their own historical terms: not as peripheral edges of empires, but as zones in their own right within a dynamic system of circulation long before modern borders were drawn.



Figure 2.1: Map of the Straits of Malacca and Singapore, published by Carey in London in 1822, with the pencil line drawn by Ambassador A.R. Falck during the negotiations for the Anglo-Dutch Treaty in London to create Dutch and British spheres of interest. Taken from the Dutch National Archives (Call No.: 4. AANW.1455). Pencil line darkened by CIL.

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Chapter 3

Hydrography of the Straits of Malacca and Singapore

o/s

Parry Oei

Introduction

The Straits of Malacca and Singapore (SOMS) forms one of the longest and critical waterways in the world. Stretching approximately 900 kilometres (485 nautical miles), it is the main link between the South China Sea and the Indian Ocean and is relied upon heavily by shipping for global trade. Navigation through the SOMS is challenging, with the high traffic density being influenced by hydrographic and oceanographic conditions.

Unlike the wide space in open seas, where vessels have room to manoeuvre and alter course freely, the SOMS confines movements within a long and narrow waterway. Over much of the length, ships must navigate within the confines and limits imposed by the waterway. The depth decreases rapidly from over 1,000 metres to mostly less than 30 metres over a stretch of about 500 kilometres (270 nautical miles), from One Fathom Bank off Port Klang to Horsburgh Lighthouse on Pedra Branca.

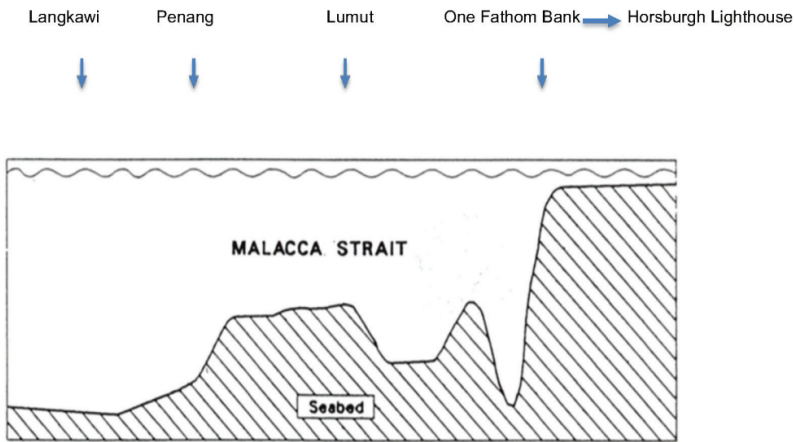


Figure 3.1: Depth of water is one of the most immediate concerns.

In several sections, depths are only marginally greater than the loaded condition draught of the largest ships that transit them. Under-keel clearance is a primary and constant concern for a ship navigating in the SOMS.

There is little room for error. A small miscalculation in position or under-keel clearance can have serious consequences.

The seabed profile adds another layer of complexity. It is not uniform or predictable. Shoals and rock outcrops interrupt otherwise navigable channels. In some areas, large sand waves exceeding 10 metres in height form and shift over time, subtly altering depths and seabed profiles. Wrecks and submerged obstructions, some historic, some more recent, lie along or near established routes.

These features are not static. The seabed in parts of the SOMS is dynamic. It is shaped by currents and sediment movement. What is safe at one point in time may not remain so indefinitely.

Current flow patterns further complicate navigation.

The SOMS is influenced by tidal flows from both the Indian Ocean and the South China Sea, as well as seasonal monsoon patterns. These forces interact to produce strong and sometimes unpredictable currents, particularly in narrower sections. For large and loaded

ships with limited manoeuvrability operating in confined waters, even moderate currents can affect steering, speed, and positioning.

All this takes place within a waterway that is heavily used by global shipping.

The SOMS is not only physically constrained but also densely trafficked. Large tankers, container ships, bulk carriers, coastal vessels, fishing boats, and other small craft share the same space. The presence of so many ships reduces the margin for manoeuvre error even further, particularly in narrow or shallow sections.

Taken together, these characteristics define the SOMS as a tightly constrained, challenging, and dynamically evolving environment. Depth limitations, dynamic seabed conditions, complex currents, and traffic interact in ways that can significantly amplify risks.

It is this combination, rather than any single factor, which makes the SOMS inherently challenging to navigate. Regular hydrographic surveys are critical to support the production of accurate and up-to-date charts of the SOMS to ensure safe navigation to sustain flow of global trade through this vital waterway.

Centuries of Charting the SOMS for East–West Trade

Charts of the SOMS are not a modern invention. As early as the 15th century during the Ming Dynasty, Malacca was a key maritime hub in Zheng He's voyages. Navigation then relied on route-based maps such as the Mao Kun map, which recorded ports, compass bearings, tides, currents, monsoon patterns, and hazards like reefs and shoals. However, these maps notably lacked depth information.

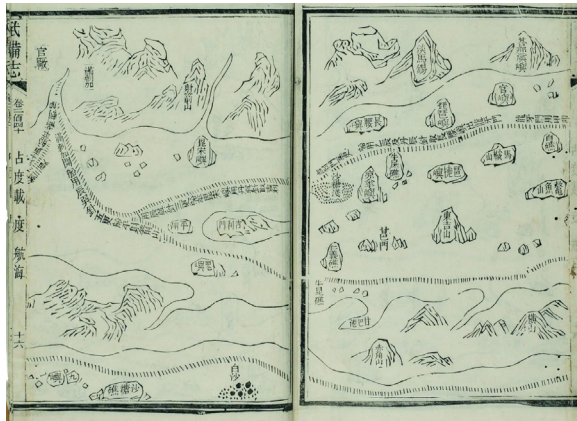


Figure 3.2: Mao Kun map.

Charts showing depth values only emerged in the 18th and 19th centuries and were often treated as strategic assets. Early surveys conducted by the English East India Company relied on lead-line sounding, a simple method using a weighted rope marked at intervals.

However, this method had clear limitations. Ships under sail depended on wind conditions, resulting in irregular zig-zag survey patterns rather than systematic coverage. Even today, some remote, non-critical areas in the SOMS still rely on depth information collected during the early 19th century, a reminder of how far hydrography has evolved.



Figure 3.3: Zig-zag pattern showing ships track. Depth soundings by lead-line.

Source: Drawing Courtesy from UKHO Archives.

The introduction of single-beam sonar in the 1960s improved survey efficiency, but gaps between survey lines remained. Near-total seabed coverage is now possible with the advent of multi-beam sonar systems. These systems provide detailed mapping of wrecks, obstructions, and seabed features, significantly enhancing navigational safety in complex waterways like the SOMS.

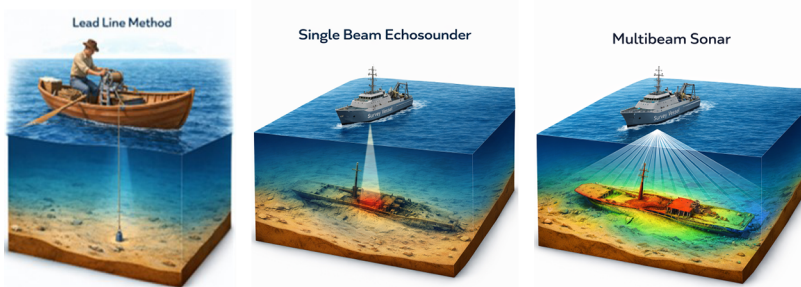


Figure 3.4: The evolution of hydrographic surveying has, in many ways, been a quiet revolution from lead-line sounding to single-beam echo sounders, and eventually to multi-beam systems, each step improving accuracy and coverage.

Co-operation Among Three Littoral States and Stakeholders: The Turning Point for Hydrography in the SOMS

1st Joint Hydrographic Survey (1970 – 1974)

By the late 1960s, existing navigational charts were no longer adequate. Increasing traffic and the introduction

of Very Large Crude Carriers and vessel with deep draught, demanded better coverage and greater precision.

In 1969, Japan, being a major user of the SOMS for oil transport, initiated a preliminary hydrographic survey. The results from the survey revealed serious safety concerns, including shallower-than-charted depths and previously unreported wrecks.

In response, the three Littoral States of Indonesia, Malaysia, Singapore, and Japan undertook the 1st Joint Hydrographic Survey from 1970 to 1974, covering critical areas from One Fathom Bank to Horsburgh Lighthouse within the SOMS.

Before the survey data could be processed, the grounding of the Japanese tanker “Showa Maru” in the SOMS occurred on 6 January 1975, and the incident underscored the urgency of the situation.

The hydrographic survey results led to the implementation of the first International Maritime Organization approved Traffic Separation Scheme (TSS) in May 1981, covering three sectors:

1. At One Fathom Bank
2. In the Singapore Strait
3. At Horsburgh Light Area

The International Hydrographic Organization (IHO) played a key role by ensuring that surveys met international standards for accuracy and consistency.

Another major outcome of the joint hydrographic survey was the introduction of Common Datum Charts that are still jointly maintained by the three Littoral States, based on unified World Geodetic System 1972 (WGS 72). This provided mariners with greater positional consistency and practical solutions to real navigational challenges in the SOMS.

The photo below shows the delegates from the three Littoral States and Japan who were instrumental in co-ordinating the Joint Hydrographic Survey, which led to the introduction and implementation of the TSS.



Figure 3.5: Delegates who attended the final meeting of the Joint Hydrographic Survey held in Tokyo in April 1971. Captain Goh Choo Keng (seated fourth from the left) was Singapore's Head of Delegation.

2nd Joint Hydrographic Survey (1996 – 1998)

Two decades later, reports of shifting sand waves, new wrecks, and shoals prompted further action. In 1993, the Tripartite Technical Experts Group (TTEG) comprising members from the three Littoral States proposed a hydrographic re-survey to extend the TSS into a continuous 500-kilometre system.

The hydrographic re-survey, conducted between 1996 and 1998, focused on critical high-risk areas within the SOMS, as shown below. The extended TSS was implemented in December 1998.

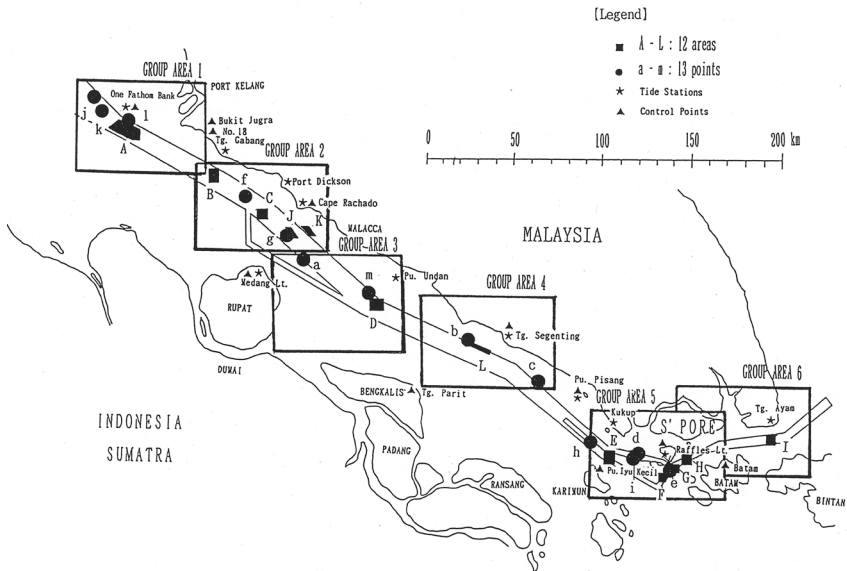


Figure 3.6: Location Map of Study Areas and Points.

The three Littoral States also updated the Common Datum Charts to the latest World Geodetic System 1984 (WGS 84), aligning charts with the Global Navigation Satellite System. This marked the beginning of a shift towards digital navigation, including the introduction of Electronic Navigational Charts (ENCs). Hydrography was no longer just about charts. It became part of an integrated digital navigation ecosystem.

3rd Joint Hydrographic Survey (2018 – 2023)

Hydrographic effort in the SOMS remains a work in progress, and this is reinforced by real-world incidents:

- 1992 – Royal Pacific collision: A cruise ship carrying 355 passengers sank after colliding with a fishing vessel. I was directly involved in the joint search operation, which located the wreck off Port Dickson.
- 1997 – Evoikos oil spill: A collision in the SOMS released 28,500 tonnes of oil and caused one of the worst oil spills in SOMS.
- 2000 – Natuna Sea grounding: The tanker ran aground after deviating from the traffic lane, highlighting navigational risks even with existing systems.

These incidents underline a simple truth: Hydrography is not just about charts, but also about saving lives, property, and protecting the marine environment.

The most recent joint hydrographic survey, conducted between 2018 and 2023 with support from Japan under the ASEAN (Association of Southeast Asian Nations)

framework, focused on areas shallower than 30 metres. The COVID-19 pandemic delayed the completion of the survey by two years. The survey results were used to update the ENC's jointly produced by the three Littoral States.

Notably, joint hydrographic surveys continued even in areas with unresolved maritime boundaries, reflecting a shared commitment to navigational safety in the SOMS.

The complexity of navigating in the SOMS is best shown by comparing the Common Datum Charts Number 200 produced in the 1981 and 2026 Editions, covering the area off One Fathom Bank in the SOMS.

The 2026 Edition shows numerous stretches of long and very shallow sand waves on both sides of the TSS. The greater details were the result of multi-beam sonars. Navigating in these waters is particularly critical for ships with draft of more than 15 metres.

The Straits of Malacca and Singapore

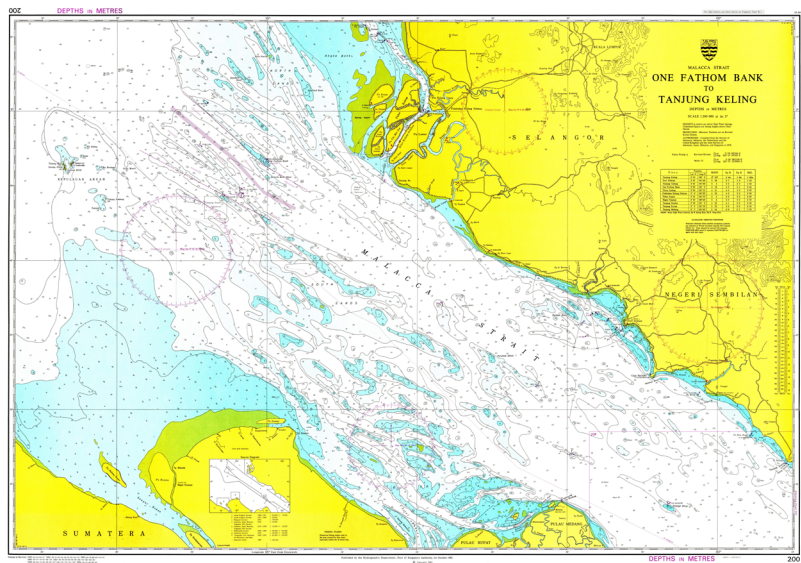


Figure 3.7: Common Datum Chart 200 (1981 Edition).

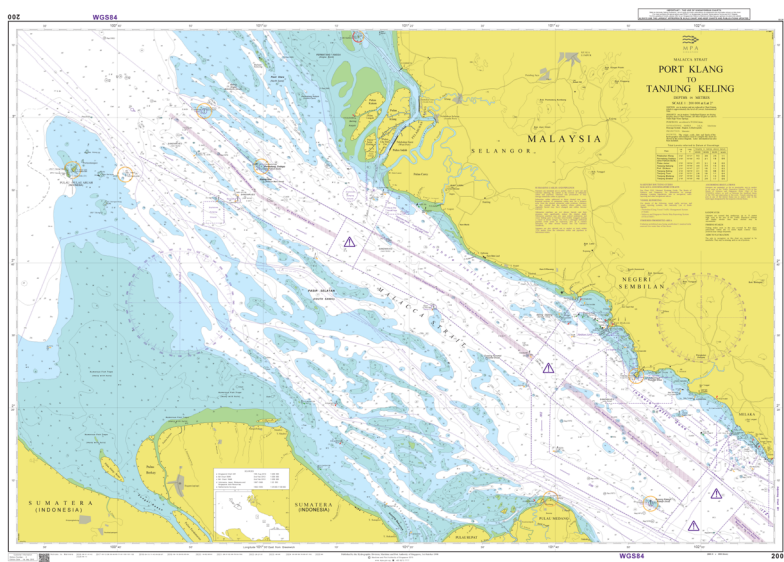


Figure 3.8: Common Datum Chart 200 (2026 Edition).

Technology and Its Limits

Even modern systems have limitations. Following the 2004 Sumatra tsunami, reports emerged of a sudden “shoal” in waters north of One Fathom Bank, previously charted at over 1,500 metres.

We investigated the reports and requested assistance from passing vessels to verify the depth in the area. It eventually became clear that the anomaly was not a real seabed feature, but a limitation of the echo sounders being used. The instruments were misinterpreting returning signals at great depth.

This served as a critical reminder: Understanding how data is generated is just as important as the data itself.

Looking Ahead: Hydrography Evolving with Technological Advances and Global Concerns

Digital Navigation

The transition from the existing to the new ENC standards will allow for richer, more flexible navigation products.

In 2024, the IHO-Singapore Innovation and Technology Laboratory successfully demonstrated Dual Fuel ENCs, i.e., existing and new products, in sea demonstrations.

With the co-operation and collaboration among the three Littoral States and Japan, the project successfully demonstrated at sea the use of Dual Fuel ENC in the SOMS, including wireless updating.



Figure 3.9: 1st Sea-Demonstration onboard Indonesian Navy Hydrographic Survey Ship, August 2024.



Figure 3.10: 2nd Sea-Demonstration onboard Singapore's Maritime and Port Authority's Hydrographic vessel, September 2024.

The completion of the sea demonstrations was timely, as the Italian Naval Training Ship “Amerigo Vespucci” had planned to transit the SOMS, and they volunteered to participate in the project. To accommodate their participation, special arrangement had to be made to install a prototype electronic chart display and information system (ECDIS) and Dual Fuel ENC's onboard. The prototype ECDIS used in the project was developed and provided by the Republic of Korea. It was an opportunity we had to seize, as it represented a multi-nation collaboration and a major step forward in digital navigation.



Figure 3.11: Onboard the Italian Naval Training Ship “Amerigo Vespucci” that used Dual Fuel ENC’s of the SOMS.

Future Geodetic Systems

The world has become overly dependent on Global Navigation Satellite System. Not knowing or using the wrong geodetic datum may risk ships running into hazards such as reefs or missing narrow channels. It is particularly dangerous in coastal navigation where waterways may be narrow and congested.

There is also ongoing discussion about transitioning from WGS 84 to the International Terrestrial Reference Frame 2020 (ITRF 2020). As global navigation systems evolve, hydrographic offices must remain prepared for future changes. We are also mindful of the fact that our Common Datum Charts covering the SOMS are presently

on WGS 84 since 1995 and should be prepared for the eventual change if needed.

Marine Environmental Protection and Climate Change

Hydrography also plays a key role in marine environmental protection. Accurate and timely updating of charts helps prevent and sometimes manage groundings and oil spills.

Regional studies, such as the ASEAN-Australian Regional Ocean Dynamics programme between 1985 and 1995, have examined tidal patterns and water movement critical for both navigation and environmental management. These studies of the SOMS are important as tidal currents are complex, and we need a better understanding of water movement to mitigate navigational risks and protect our marine environment and coastline.

Conclusion

Hydrography in the SOMS is a story of continuous adaptation, collaboration, and shared responsibility. From early route-based maps to advanced digital systems, each phase has responded to the needs of one

of the world's most complex and heavily trafficked waterways.

While technology continues to advance, it is co-operation among the Littoral States and the global maritime community that remains the foundation of navigational safety.

Hydrographic surveying is never a completed task. It is an ongoing commitment to accuracy, safety, and the stewardship of these vital waters. The work, as always, is to stay ahead of these changes.

Chapter 4

Commercial and Strategic Importance of the SOMS

✎

Mary Seet-Cheng

Even in today's fast-paced, technologically advanced world, 80 per cent of the world's goods still get transported by sea because it remains the most efficient and cost-effective way to bring goods to market. These goods get loaded on vessels for often long maritime voyages to their destination. Most times the vessels can sail through open waters, but for some parts of the journey, they must pass through narrow maritime highways as these are the shortest and safest route possible connecting major oceans and seas. These narrow maritime highways are variously referred to as critical sea lines of communication, critical maritime chokepoints, or vital shipping lanes. They refer to a geographically constricted maritime highway that

carries a high volume of vessel traffic and where any disruption of traffic flows can cause major global supply chain bottlenecks, drive up shipping costs, energy prices, and create geopolitical tensions.

In 2004, the International Maritime Organization (the United Nations specialised agency responsible for regulating international shipping) launched an initiative called the “Protection of Vital Shipping Lanes”. It focused the maritime world on the need to enhance safety, security, and environmental protection in critical maritime shipping lanes. The initiative specifically focused on the Straits of Malacca and Singapore (SOMS) with good reason. The SOMS is considered the most vital economic and strategic shipping lane in the world. The 2004 IMO document said that “more than half of the world’s annual merchant tonnage passes through the SOMS, with the majority continuing on to the South China Sea”. It also pointed out that “oil flows through the SOMS into S. China Sea was three times greater than the Suez Canal and 15 times greater than oil flows through the Panama Canal”. It added that “if for some reason the SOMS were closed, nearly half of the world’s fleet would be required to sail some 560 miles further, resulting in the prolongation of voyage times and generating substantial increase in the requirement for

vessel capacity”. Further, “closure of the SOMS could be expected to immediately raise freight rates worldwide”. The importance of the SOMS has grown much since 2004 for several reasons.

Strategic Location of SOMS

The SOMS is approximately 900 kilometres long and is in Southeast Asia. At its narrowest point, it is just 1.7 miles wide. It is bordered by the three littoral States of Indonesia, Malaysia, and Singapore. Its location, connecting the Indian Ocean to the South China Sea and beyond to the Pacific Ocean, makes it strategically important as it facilitates trade flows between Europe, Africa, the Middle East, and the Indian Subcontinent with ASEAN (Association of Southeast Asian Nations), China, Japan, and South Korea and beyond. Being a through-traffic waterway, vessels can sail on through to many more destinations between the East and the West. (As compared to the Strait of Hormuz, which connects the enclosed Persian Gulf to the Arabian Sea). Because of this strategic location, SOMS has become an even more vital shipping lane for oil flows from the Middle East to Asia. In the 2004 IMO document, oil flows through the SOMS were stated as 10.3 mbpd (million

barrels per day) in 2002. It was predicted that demand could double these flows over the next two decades. This prediction was spot on, as the current figure through the SOMS stands at 23.2 mbpd.

Wide Range of Cargo Flows through the SOMS

The blockade of the Strait of Hormuz, precipitated by the war between US/Israel and Iran in February 2026, created a major fuel crisis because it is a vital shipping lane carrying 20 per cent of the world's energy flows. If such an event happened in SOMS, there would be an even wider impact beyond energy security. If SOMS were blocked, as indicated in the IMO document, nearly half of the world's shipping fleet would be required to reroute around the Indonesian Archipelago through the Lombok or Sunda Straits. However, these alternative routes could add several days to sailing times, tie up global shipping capacity, add to shipping costs, and affect energy prices. But it would not just be the oil flow that would be affected. A wider range of cargo such as raw materials, commodities, manufactured goods, and food are carried through the SOMS. It is reflected by the greater range of vessel types that transit through the SOMS. The

biggest users of the SOMS are container ships and bulk carriers, not oil tankers. Using data published by Singapore's Maritime and Port Authority on vessel arrivals at its port as a proxy, in 2025, container ships and bulk carriers brought in two-thirds of the tonnage. The oil tankers brought in the other one-third. Any closure of the SOMS will have a major impact on all kinds of global supply chains, not just on energy supply chains.

Serving the Rapidly Growing Economies of Southeast and East Asia, Particularly China

When IMO presented the details of traffic through the SOMS in its 2004 document, it recorded that there were 60,000 vessel transits that year. In 2025, the figure was up to 120,000 transits. This doubling of traffic in just over two decades reflects the shifting of the world economic centre of gravity to Asia. Figures from the International Monetary Fund (IMF) showed that China's share of global GDP was 4 to 5 per cent in 2004. In 2025, it was 15 to 16 per cent. China had become the factory of the world. China's demand for oil has tripled during this period, in tandem with its industrial expansion. For

ASEAN countries, the IMF data for 2004 was 2.3 per cent of global GDP. In 2025, it rose to 7.4 per cent, making the group the fifth largest economy in the world. Indonesia's economy contributes 35 per cent of this total combined GDP of the ASEAN bloc, making it the largest economy in Southeast Asia.

Importance of SOMS to China, Japan, and South Korea

The SOMS is a critical maritime highway for China. Eighty per cent of its oil imports flow through this waterway on their way to China from the Middle East and Africa, as with a vast portion of its trade with the world (estimated at 60 per cent). As the primary conduit between the Indian Ocean and the South China Sea, it is central to China's energy security, economic growth, and strategic planning. This dependence has been labelled as the "Malacca Dilemma", as any blockade of this highway could cripple China's economy. Its reliance on SOMS has been viewed as a major strategic weakness. It has been trying to develop alternative routes to bypass the SOMS, such as the China–Myanmar oil pipelines, and other overland routes. However, sea transport remains the most efficient and cost effective. The SOMS

is the shortest and most reliable transport route, thus making its safety and continuous access essential to China's national security. China has been an active participant in co-operative activities among the littoral States and user States to keep the SOMS open and safe for maritime transit.

The SOMS is similarly a critical maritime lifeline for Japan. Approximately 90 per cent of its crude oil imports from the Middle East transit through this waterway. As it lacks its own domestic energy resources, it depends on energy imports. The SOMS is also the primary route for Japan's trade with the West. Any blockage of this route will put its economy and national security at risk. Therefore, Japan has worked with the three littoral States of Indonesia, Malaysia, and Singapore since the 1960s, to ensure safety of navigation in the SOMS by providing financial and technical aid to reduce navigational risks and improve maritime security, especially when piracy and ship robberies in SOMS became an issue. It has continued to be an active member of all co-operative activities among the littoral States, users, and other industry stakeholders to keep the SOMS open and safe for maritime transit.

The Republic of Korea (ROK) is similarly reliant on the SOMS for facilitating over 80 per cent of its crude oil imports from the Middle East. As the shortest link to Europe, the SOMS is a vital maritime artery for its manufactured exports and imports of raw materials. As a major user State of the SOMS, the ROK relies heavily on the continued safety and efficiency of navigation through the SOMS. The ROK hence plays an active role in supporting the littoral States' efforts to enhance navigational safety and marine environment protection in the SOMS, including by making regular financial contributions towards the renewal and maintenance of critical aids to navigation in the SOMS and contributing to capacity building in the three littoral States.

Importance of SOMS to Singapore, Malaysia, and Indonesia

In September 2007, at the inaugural Singapore Maritime Lecture during Singapore Maritime Week, then Minister Mentor Lee Kuan Yew declared that “Singapore's raison d'être was its port”. Indeed, Singapore often traces its modern beginnings to its establishment as a British entrepôt port in 1819 by Sir Stamford Raffles to counter Dutch dominance in the region and to protect Britain's

then crucial maritime trade route between India and China. Raffles recognised Singapore's valuable location astride the vital shipping lane of the SOMS. After independence, Singapore continued to leverage this beginning to grow into a major hub port and international maritime centre. There is no question in Singapore of the strategic importance of the SOMS to its continued economic development and prosperity. The maritime sector now contributes 6 per cent to Singapore GDP and 120,000 jobs. Its position as a major transshipment port, handling almost 45 million TEUs in 2025, has also enabled it to carve out an important role as an international logistics and supply chain hub.

Singapore has also leveraged its strategic location along the oil shipping route to attract oil majors to invest and develop it into one of the top three global oil refining and trading hubs. The oil refining and petrochemical cluster on Jurong Island accounts for 3 to 5 per cent of total GDP, supporting 27,000 skilled workers. It is now a major exporter of refined products to the region and as far as Australia, New Zealand, and the Pacific. According to Prime Minister Lawrence Wong, Singapore's role as a key node in global energy flows gives the city-state a significant advantage in energy security. The

world's leading energy companies operate in Singapore and can draw on diverse supply networks to secure alternative fuel supplies during crises.

For Malaysia, the waterway boosts Malaysia's economy via port activity, trade, and regional development. It has substantially grown its port infrastructure to benefit from the high shipping traffic through the SOMS. Container throughput through the two major ports of Port Klang and Tanjung Pelepas have reached close to 30 million TEUs, more than double since 2008. There are big plans for expansion in Port Klang, including projects at Westport and Carey Island, which would boost overall capacity to approximately 60 million TEUs by around 2040s onwards.

Compared to Singapore and Malaysia, Indonesia's port infrastructure may not be as developed to take advantage of the high vessel traffic to provide transshipment and other maritime services. However, the SOMS serves as the primary artery for the large volumes of raw materials and commodities, particularly coal, natural gas, and crude oil that it exports to major consumers like China and Japan. The high frequency of shipping services through the SOMS also dramatically reduces transit

times and freight costs for Indonesian manufacturers and agricultural producers. Conversely, the competitive shipping freight rates keep costs down for consumers and companies that import into Indonesia. Indonesia's strategic position along the SOMS also allows it to capitalise on attracting foreign investments along its coastlines to establish logistics, manufacturing and processing hubs that directly serve the passing global trade. The rapid transformation of the Riau Islands (the Batam-Bintan-Karimun corridor) into a hub for green energy, advanced technology, and infrastructure is a case in point. Its strategic position along key maritime trade routes, especially the SOMS, has facilitated Indonesia's remarkable economic growth over the last three decades, which has averaged 5 per cent per annum.

With the shift of economic centre of gravity to Asia, the importance of SOMS as a major maritime highway can only grow. Indonesia, together with the other two littoral States of Malaysia and Singapore, has the important shared responsibility to ensure it remains a safe, secure and open freeway and not a chokepoint for the flow of international trade. The three littoral States have done remarkably well in their co-ordination and co-operation over the years, beginning with the establishment of the

Tripartite Technical Experts Group (TTEG) in 1975. The TTEG has worked to implement safety of navigation measures like the Traffic Separation Scheme and the compulsory shipping reporting system STRAITREP that now guides shipping safely through the narrowest passageways of the SOMS. In 2007, in co-ordination with the IMO, the Co-operation Forum was created to enhance co-operation between littoral States and user States, and with other stakeholders, in fulfilment of Article 43 of the United Nations Conference on the Law of the Sea that called for such co-operation in straits that are used for international navigation. To address maritime security concerns such as piracy and potential acts of terrorism, the littoral States have also co-operated among their navies to implement joint naval patrols and eyes-in-the-sky surveillance systems. Through these transparent and open co-operation initiatives, the littoral States have been able to stave off external interference in the guise of helping protect maritime safety and security in SOMS. As the SOMS grows in economic and strategic importance, it is even more critical to ensure that the three littoral States remain in charge by sustaining this co-operation. This is important to keep out big power rivalry from this most vital shipping lane.



Figure 4.1: Ambassador Mary Seet-Cheng with then Secretary-General of the International Maritime Organization Lim Kitack (greeting her, back to the camera) at the IMO General Assembly 33 in 2023.

Suggested Reading

IMO Document C93/15, 7 October 2004, Protection of Vital Shipping Lanes, Note by the Secretary-General.

Prime Minister Lawrence Wong in Labour Day Speech, 1 May 2026.

Chapter 5

The UN Convention on the Law of the Sea and the Straits of Malacca and Singapore

Tommy Koh

Introduction

The straits of the world are important shipping lanes. They are governed by international law and Part III of the United Nations Convention on the Law of the Sea (UNCLOS). The International Maritime Organization (IMO), as the UN specialised agency for international shipping, also plays an important role. This chapter describes the development of the legal regime governing straits used for international navigation, Singapore's role in the negotiations and the implications of the agreement for the Straits of Malacca and Singapore.

New Nations Challenge Old Law

One important aspect of the law of the sea is the breadth of the territorial sea. By custom, coastal states were allowed to claim a territorial sea of three nautical miles within which the coastal states have sovereignty. However, the coastal states had to allow ships of all countries to pass through their territorial seas, which under international law is known as the right of “innocent passage”. A ship, exercising the right of innocent passage cannot stop in its passage and must not pose a threat to the security of the coastal state.

If a strait was wider than six nautical miles, and the territorial sea was three miles in breadth, the zone in the middle of the strait would be the high seas. The high seas corridor in a strait is very important to shipping and to the navies of the world, as coastal states have no legal right to regulate shipping in the high seas.

The end of the Second World War in 1945 brought about many revolutionary changes in the world. Most of the colonies in Asia, Africa and elsewhere became independent countries. The newly independent countries, with the support of the Latin-American countries, were unhappy with some aspects of the

existing legal order. For example, they were not satisfied with the three-nautical-mile limit for the territorial sea. Consequently, individual countries started to unilaterally claim territorial seas of six and 12 nautical miles, which resulted in much legal uncertainty.

The First and Second UN Conferences on the Law of the Sea

In 1958, the UN convened the First UN Conference on the Law of the Sea. The agenda was to adopt new conventions to govern different aspects of law of the sea, including a new Convention on the territorial sea. While the conference succeeded in adopting the 1958 Convention on the Territorial Sea and Contiguous Zone, it failed to specify the breadth of the territorial sea because participating states could not reach consensus.

In 1960, the UN convened the Second UN Conference on the Law of the Sea. The conference also could not agree on the breadth of the territorial sea. Indeed, a proposal by Canada for a six-nautical-mile territorial sea failed to pass by one vote.

In subsequent years, the legal order deteriorated significantly as some coastal states claimed territorial seas of 50 and even 100 nautical miles. The diversity in unilateral claims resulted in a state of legal chaos. The shipping community and the navies of the world were denied the clarity and certainty that they needed. It thus became imperative to arrive at an agreement on the breadth of the territorial sea.

The Third UN Conference on the Law of the Sea

In 1970, the UN decided to convene the Third UN Conference on the Law of the Sea. The ambition was to adopt a comprehensive convention covering all aspects of the law of the sea.

The breadth of the territorial sea was one of the critical issues that had to be resolved. In 1974, 51 states had claimed a territorial sea of 12 nautical miles and only 25 states claimed a territorial sea of six nautical miles. The trend was unmistakable. The majority of the coastal states of the world favoured 12 nautical miles as the breadth of the territorial sea.

The extension of the territorial sea, from three to 12 nautical miles, had a major impact on many straits. There are 116 straits used for international navigation, including the Straits of Malacca and Singapore, which are between 6 miles and 24 miles in breadth.

When the territorial sea was three nautical miles, there were high sea corridors in those straits. However, when the territorial sea was expanded to 12 nautical miles, the high seas corridors were gone. The regime of freedom of navigation in the former high seas corridors would become innocent passage.

The then two superpowers, the United States and the Soviet Union, and the other maritime powers, demanded a quid pro quo. They would accept a 12-nautical-mile territorial sea provided there was a special regime for the passage of ships, submarines, and aircraft, through, under, and over straits used for international navigation.

This issue was a conference breaker and presented significant challenges and difficulties at the conference. There was a Strait Group at the conference comprising Cyprus, Greece, Indonesia, Malaysia, Morocco, Oman, the Philippines, Spain and Yemen. Several of the important strait states refused to compromise. They

insisted that their sovereignty over their territorial sea must prevail. They argued that the superpowers and maritime powers should accept innocent passage for their ships, submarines, and aircraft. At this juncture, it may be useful for me to discuss briefly the role that the Singapore delegation played in the Third UN Conference on the Law of the Sea.

Singapore's Interests at the Third UN Conference

Singapore owes its existence to the sea. It is situated at one end of the Strait of Malacca. Singapore has a deep seaport that has historically serviced ships going from the West to the East and from the East to the West. It has been an entrepôt port and a trading hub. As a result, Singapore has championed free trade and freedom of navigation.

Singapore became independent in 1965 and joined the UN in the same year. In 1968, I was appointed Singapore's Permanent Representative to the UN. When the UN convened the Third UN Conference on the Law of the Sea in 1970, I convinced Singapore that we should

participate in the Preparatory Committee and, subsequently, in the conference.

When the conference began, I had returned to the UN as Singapore's Permanent Representative. I was the leader of the Singapore delegation to the conference. In the beginning, there were only three of us: S. Jayakumar, Chao Hick Tin and me. S. Tiwari subsequently joined our small team.

Singapore's Agenda

What was Singapore's agenda at the conference? At the highest level, we wanted to help the conference adopt a new convention that would be comprehensive, just, and forward-looking. Second, we wanted to protect the freedom of navigation, which is so important to shipping and international trade. Third, we wanted to protect our interests as a port state. Fourth, we wanted to protect our right to gain access to the high seas for our ships and aircraft. Fifth, we were prepared to play the role of a consensus-maker and bridge builder.

In the beginning, the conference was dominated by the coastal states, especially the Latin-American coastal

states. They were united and co-ordinated in their efforts to advance their positions. States that had extensive territorial sea claims, in particular, were strongly opposed to the positions of delegations like Singapore, which prioritised freedom of navigation.

Group of Landlocked and Geographically Disadvantaged States

We decided to do something about the situation. Singapore and Austria formed the group of Landlocked and Geographically Disadvantaged States. The group had 55 members, comprising 29 landlocked countries and 26 geographically disadvantaged states. The size of the group gave us leverage and influence at the conference.

Battle at the UN General Assembly

We managed to demonstrate our power at the 27th UN General Assembly. The Singapore delegation, led by Professor Jayakumar, tabled a draft resolution asking the Secretary-General to prepare a study on the implications of the various claims put forward by the

coastal states. The coastal states opposed the draft resolution, and several jointly tabled a “killer” amendment to the draft resolution. Fortunately for us, the amendment failed to be adopted by one vote, and the draft resolution was passed by a vote of 69 in favour, 15 against and 41 abstentions. After this episode, there was increased respect for the Singapore delegation.

In the negotiations of the Straits regime, Singapore recognised that as a littoral State in the Straits of Malacca and Singapore, it was important to have rules for the safety of navigation and prevention of pollution. Ensuring unimpeded navigation in the Straits of Malacca and Singapore was also critical. Singapore played an active role in the negotiations, brokering a compromise between the maritime powers and the Straits States.

Fiji and UK to the Rescue

To break the impasse on navigation in straits, Fiji and the UK decided to make a joint initiative. They convened an informal negotiating group on Straits Used For International Navigation. I should explain that the complexity of the agenda and the difficulty of arriving at a consensus caused several conference leaders to

convene informal negotiating groups to tackle some of the most difficult issues such as the Exclusive Economic Zone and the system of dispute settlement. It was these small informal negotiating groups that helped the conference to solve the most difficult issues. The Singapore delegation was invited to participate in all the privately convened negotiating groups.

Fiji and UK co-chaired the group on Straits. They decided not to include the two superpowers in the group and to exclude Straits States, which were opposed to finding a compromise.

The 13 delegations that were invited to join the group were: Argentina, Australia, Bahrain, Bulgaria, Denmark, Iceland, India, Italy, Kenya, Nigeria, Singapore, the United Arab Emirates, and Venezuela. The composition of the group was inclusive and included countries from the different continents. In my view, the co-chairs also picked delegations with moderate views and had an influence at the conference.

Success

The group was established at the conference in 1975. We worked hard and under the able leadership of Fiji and the UK, and we succeeded in adopting a draft text on Straits Used For International Navigation. We submitted our paper to the chairman of the conference's Second Committee, who incorporated it into the Informal Single Negotiations Text (ISNT).

When the ISNT was published, the reaction of most of the delegations was positive. They thought the text had struck the right balance between the rights of the coastal states and the rights of the international community. Naturally, the delegates which were opposed to compromise objected to the text. However, they represented the minority view, and the majority prevailed.

Most coastal states supported the compromise because they wanted agreement for a 12-nautical-mile territorial sea. Many attempts were made by other states to delete the negotiated text, but they were unsuccessful. The articles in what became Part III on the regime of straits used for international law of the UN Convention on the

Law of the Sea, ultimately came from the Fiji-UK informal negotiating group.

The compromise included a new “transit regime” of passage for straits that were not otherwise subject to innocent passage or a regime specified under a longstanding international convention.

A Strait Used For International Navigation under UNCLOS?

Building on the famous judgment of the International Court of Justice in the Corfu Channel case (United Kingdom v Albania), Article 37 of UNCLOS states that the transit passage regime applies to straits that are used for international navigation between one part of the high seas or exclusive economic zone and another part of the high seas or an exclusive economic zone.

The Straits of Malacca and Singapore qualify as Straits Used For International Navigation subject to the transit passage regime. Some other straits that are subject to transit passage are Gibraltar, Dover, Hormuz, and Bab-Al-Mandeb.

What is Transit Passage?

I should explain that “transit passage” is a new concept in the law of the sea. In the hierarchy of navigational freedoms, it is a stronger regime than innocent passage. It is, however, weaker than the high seas’ freedoms of navigation and overflight.

Transit passage means “the freedom of navigation and overflight solely for the purpose of continuous and expeditious transit of the straits.”

The term, “freedom of navigation and overflight” is legally significant because it is language generally associated with the high seas. However, unlike the high seas, the transit must be continuous and expeditious.

The regime of transit passage has the following characteristics:

- i. It applies to all ships, including warships.
- ii. Submarines may remain submerged during their passage; and

- iii. A ship, submarine, and aircraft enjoys unimpeded passage through, under, and over a Strait Used For International Navigation.

Singapore Exception

There is, however, a Singapore exception. A ship or aircraft transiting the Straits of Malacca and Singapore, may stop in Singapore and still enjoy the regime of transit passage.

Interpretation

I want to make a few comments. First, the regime of transit passage applies to all ships, including warships. What this means is that Indonesia, Malaysia, and Singapore must accept that warships have the same right as other ships, in transiting the Straits of Malacca and Singapore. Second, the Straits States cannot require submarines to come to the surface in transiting the Straits, whereas in innocent passage, submarines must navigate on the surface. This is very important to the great powers because they do not want their adversaries to know the location of their submarines. Third, the

Straits States cannot impede the passage of ships, submarines and aircraft, through, under, and over the Straits. Fourth, there is a right of transit overflight of straits, which does not apply for innocent passage.

Duties of Ships and Aircraft during Transit Passage

The convention imposes a number of duties on ships and aircraft during transit passage. First, the ship or aircraft must proceed without delay through, under, or over the strait. A ship cannot stop to fish or to conduct research. Second, a ship or aircraft in transit must not use force, or threaten to use force, against the sovereignty, territorial integrity, or political independence of the state bordering the strait. I think the meaning of this duty is clear and does not need elaboration. Third, a ship or aircraft in transit must not undertake any activity, other than expeditious passage, except when they face “force majeure” or distress. The term, “force majeure” means extraordinary and uncontrollable events, such as, natural disasters, war, or pandemics. The term “distress” would include situations such as fire on the ship, a crew member who is critically ill, or a passenger who has fallen into the sea.

Compulsory Pilotage

The UNCLOS came into force in 1994. Since then, the regime of transit passage has been observed both by the Straits States and by user States. The only controversy that has occurred resulted from a unilateral action by Australia and Papua New Guinea in 2009.

The Torres Strait

The Torres Strait lies between Australia and Papua New Guinea. In 2003, the two countries submitted a proposal to the IMO. The proposal would extend the Great Barrier Reef Particularly Sensitive Sea Area, to the Torres Strait, and also extends the compulsory pilotage system to the Torres Strait.

The Torres Strait is a Strait Used For International Navigation. The question is whether Australia, a State bordering a strait, has the legal power, consistent with the transit passage regime under UNCLOS, to impose a system of compulsory pilotage.

The question was debated at the IMO. In the discussion, the United States opposed the Australian/Papua New Guinea proposal. Singapore joined several other countries in supporting the US view that there was no international legal basis for mandatory pilotage for ships in transit in any Strait Used For International Navigation.

Bilateral Consultations

Singapore held two rounds of bilateral consultations with Australia to resolve the disagreement amicably. The first round was held in 2007.

The second round was held in Singapore in March 2009. I led the Singapore delegation at the second round. Although the talks were cordial and friendly, they were unsuccessful. Australia agreed not to claim that it had the support of the IMO. However, it was determined to impose compulsory pilotage on ships transiting the Torres Strait.

Compulsory Pilotage in the Torres Strait

Although the proposal of Australia/Papua New Guinea was not accepted by the IMO and was opposed by the US, Singapore, and other delegations, Australia proceeded to issue a Marine Notice in 2009 to implement the system of compulsory pilotage for ships transiting the Torres Strait.

A Bad Precedent

The proposal of Australia/Papua New Guinea is inconsistent with the right of transit passage in Straits Used For International Navigation as enshrined in UNCLOS. Australia's action is disappointing because Australia was a member of the Fiji-UK informal negotiating group that drafted the chapter on Straits. Fortunately, no other Strait State to date has adopted similar measures, suggesting that the system of compulsory pilotage in the Torres Strait has not gained acceptance and remains, if anything, a bad precedent that should not be followed.

No Toll is Allowed

I will now deal briefly with whether a Strait State, such as Iran, can impose a toll on ships transiting a Strait Used For International Navigation. The answer is that, under UNCLOS, a Strait State cannot do so. The power of a Strait State to make laws and regulations relating to transit passage is set out in Article 42 of UNCLOS. The Article spells out in Article 42, para 1, and sub-paras (a), (b), (c) and (d), the four areas in which such laws and regulations may be adopted by the Strait State. Imposing tolls on ships in transit passage is not included in the article.

Conclusion

The Third UN Conference on the Law of the Sea was not merely a codification conference. The conference was innovative and invented several new concepts in the law of the sea and international law, such as, the Exclusive Economic Zone, Archipelagic State, Transit Passage, Archipelagic Sea Lanes Passage, and the Common Heritage of Mankind. The Third Conference fixed the breadth of the territorial sea at 12 nautical miles, which the First and Second Conferences had failed to do. This chapter explains how the conference

came to adopt the new regime of “transit passage” for Straits Used For International Navigation, and the role which Singapore played in the negotiations.



Figure 5.1: (L to R) Emeritus Professor Robert Beckman, Professor S. Jayakumar, Ambassador Satya Nandan, and Ambassador Tommy Koh. It was taken at the NUS Centre for International Law. Ambassador Nandan had been invited to speak about his experience at the United Nations Conference on the Law of the Sea.

Suggested Reading

1982 United Nations Convention on the Law of the Sea.

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Chapter 6

The Traffic Separation Scheme and Co-operation between the Littoral States and the International Maritime Organization

✎

Ang Wee Keong

For centuries, the fates of the three littoral States – Indonesia, Malaysia, and Singapore – have been tied to the Straits of Malacca and Singapore (SOMS). Home to thriving ports and coastal towns, the SOMS has helped to shape the histories and identities of the littoral States, facilitating a rich exchange of goods and cultures among its peoples. The SOMS is also a vital economic lifeline for the littoral States and of great commercial and strategic significance to other users of

the SOMS. The SOMS serves as the shortest, single, and contiguous sea trade route connecting the Indian and the Pacific Oceans, physically linking the markets in the East and the West. As one of the busiest shipping lanes in the world, over 100,000 vessels, carrying a third of global container and a quarter of global seaborne oil trade, transit the SOMS each year.

Given its importance to the global economy, ensuring freedom of passage through the SOMS, both in principle and practice, is of existential importance to the littoral States and their users. As a Strait Used for International Navigation (SUIN), vessels enjoy the right of transit passage – a fundamental right enshrined in the 1982 United Nations Convention on the Law of the Sea and reflected in customary international law. Vessels do not pay fees as a requirement to transit the SOMS, nor are they subject to compulsory pilotage.

Despite the high (and increasing) volume of vessel traffic in the SOMS, trade continues to flow through the SOMS safely and smoothly, and few marine incidents affecting traffic movement occur. This begs the question: How has the SOMS remained open and safe for international maritime passage over all these years?

Need to Enhance Navigational Safety in the SOMS

Commercial vessels did not always enjoy the same level of navigational safety in the SOMS as they do today. The conclusion of the Second World War ushered in an era of rapid post-war economic recovery and industrialisation. This precipitated a sharp rise in global energy demand in the late 1950s and 1960s. The increased demand for oil shipments consequently led to the introduction of Very Large Crude Carriers (VLCCs) – large oil tankers able to carry over 200,000 deadweight tonnes. These vessels were designed to transport oil more efficiently over long distances but had deeper draughts and were less manoeuvrable.

The navigational safety challenges posed by VLCCs was brought into sharp focus in the late 1960s, following the occurrence of several high-profile marine incidents involving large oil tankers. On 18 March 1967, over 100,000 tonnes of crude oil was spilt after *TORREY CANYON* collided with rocks off the English Channel. The SOMS was similarly not spared. Less than three weeks after the *TORREY CANYON* oil spill, *TOKYO MARU* ran aground off the coast of Medan, Indonesia.

That same year, *IDEMITSU MARU* – the very first VLCC to transit the SOMS – scraped the seabed twice when transiting the SOMS. The littoral States and users of the SOMS were fortunate that both incidents resulted in minimal damage. However, they highlighted the significant risks posed by shallow and danger areas on vessels transiting the SOMS that were unmarked on existing charts. The warning was loud and clear: urgently improve navigational safety in the SOMS or risk further (and greater) mishaps.

Soon after the *TORREY CANYON* oil spill, then the largest oil spill ever recorded, discussions on the need to improve navigational safety and marine environment protection took place at the Inter-Governmental Maritime Consultative Organization (IMCO; then-IMO). Of note, there were calls for the establishment of a traffic separation scheme (TSS), akin to a marine highway with traffic lanes, to improve navigational safety in the SOMS. It was agreed that prior to the introduction of a TSS, there was a need to first survey the SOMS. New aids to navigation (AtoNs) along the SOMS were also required to enhance navigational safety. The littoral States would separately study the proposed establishment of a TSS in the SOMS before returning with a joint proposal at the IMCO.

Establishment of the Tripartite Technical Experts Group

So began work to enhance navigational safety in the SOMS. As a major user State of the SOMS and one of the world's largest shipbuilding nations, Japan generously offered to support the littoral States' efforts through its establishment of the Malacca Strait Council (MSC) in 1969. Following the conduct of a preliminary survey in 1969, the 1st Joint Hydrographic Survey was carried out by the littoral States and MSC from 1970 to 1974. This enabled the littoral States to identify serious navigational dangers and update the nautical charts used by mariners to transit the SOMS. Between 1969 and 1971, MSC also funded and supported the installation of several AtoNs along the SOMS, including beacons, buoys, and lighthouses.

As these efforts were underway, the governments of the three littoral States convened the 1st Tripartite Ministerial Meeting in October 1971 to develop a common position on matters relating to the SOMS. Notwithstanding legal and political differences concerning the "internationalisation" of the SOMS, the three littoral States recognised the urgent need to

enhance navigational safety in the SOMS. Among other matters, they agreed on the need for tripartite co-operation on navigational safety in the SOMS and to establish a body for co-operation to co-ordinate efforts related to navigational safety.

The grounding of *SHOWA MARU* off Pulau Sebarok, Singapore on 6 January 1975 further emphasised the need for tripartite co-operation to improve navigational safety and marine environment protection in the SOMS. The incident resulted in what was back then the worst oil spill in the SOMS, spilling over 3,300 tonnes of crude oil and racking up more than US\$100 million worth of damage claims. With the oil spill fresh in their minds, the littoral States agreed to establish an “expert group” during the 2nd Tripartite Ministerial Meeting in February 1975, later known as the Tripartite Technical Experts Group on the Safety of Navigation in the SOMS (TTEG), to study possible measures to address the navigational safety risks posed by VLCCs transiting the SOMS. Of note, the littoral States agreed to the establishment of a TSS in the SOMS, marking a pivotal moment in their tripartite co-operation.

Adoption of the Traffic Separation Scheme

The technical experts from the three littoral States worked closely on what the TTEG had set out to do. They designed the proposed TSS and the rules that deep draught vessels and VLCCs transiting the SOMS had to comply with, such as designated routes, speed limits, and under-keel clearance. In doing so, the TTEG, alongside MSC, studied the vessel traffic patterns, hydrographic conditions, and navigational risks of the SOMS, incorporating findings from the 1st Joint Hydrographic Survey.

The work of the TTEG culminated in the three littoral States signing the “Agreement on Safety of Navigation in the SOMS” during the 3rd Ministerial Meeting in February 1977. The Foreign Ministers of the littoral States approved the TTEG’s proposed TSS design, which comprised deep-water routes for deep draught vessels, a co-ordinated system of AtoNs and a TSS in three areas: One Fathom Bank; the Main Strait and Philip Channel; and off Horsburgh Lighthouse. The proposed TSS and associated “Rules for Vessels Navigating through the SOMS” were jointly tabled by the three littoral States at the IMCO and adopted in November 1977, before coming into force on 1 July 1981.

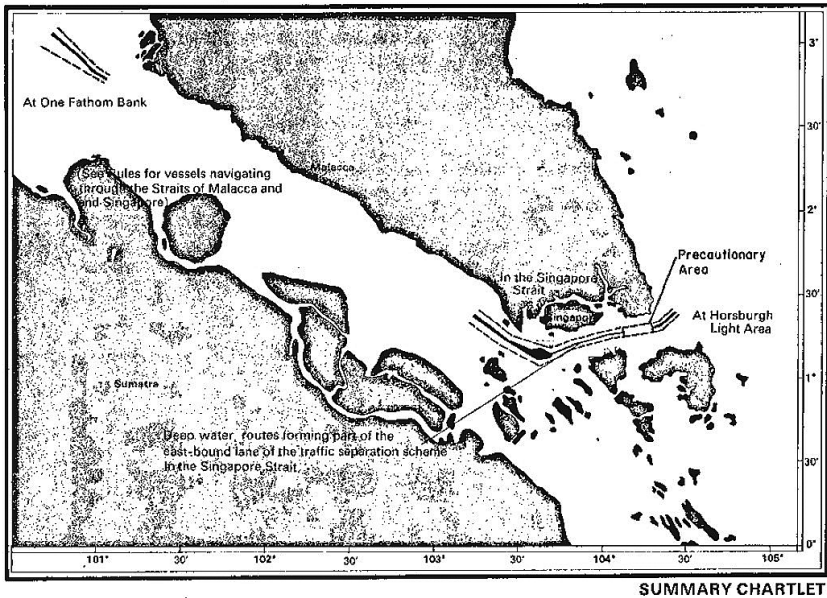


Figure 6.1: Traffic Separation Scheme in the SOMS (1981).
Source: Singaporean Notices to Mariners (SNM), Edition No. 2/81.

While the TTEG achieved what it was initially tasked with, there remained work to be done. The three littoral States' successful establishment of the TSS demonstrated the tangible value of strong, technical cooperation. It also proved the effectiveness of the TTEG as an operational platform for the three littoral States to collectively improve navigational safety and marine environment protection in the SOMS. The littoral States agreed for the TTEG to continue meeting annually, to continue engaging users of the SOMS on matters relating

to navigational safety and marine environment protection, and for decisions at the TTEG to continue being made based on consensus.

Continuation of the TTEG and Extension of the Traffic Separation Scheme

A significant increase in traffic volume in the SOMS and a rising occurrence of marine incidents between the 1980s and early 1990s, reaffirmed the littoral States' belief in the continued need for the TTEG. The occurrence of notable incidents in 1992, such as the collision between *USS INGERSOLL* and *MATSUMI MARU*, and the *NAGASKI SPIRIT* oil spill – which led to 12,000 tonnes of oil being spilt – prompted a review of the existing measures in the SOMS. During the 18th TTEG meeting in February 1993, the three littoral States discussed the possibility of extending the TSS to cover the areas between Raleigh Shoal and Tanjung Piai. The meeting also discussed the proposed establishment of a mandatory ship reporting system in the SOMS – later known as *STRAITREP* – and the provision of Vessel Traffic Services (VTS), where shore-based authorities could provide navigational advisories to assist onboard decision-making.

Soon after, the TTEG commenced work to review the TSS and establish STRAITREP. In May 1998, the IMO adopted the littoral States' proposal to extend the TSS to cover over 260 nautical miles between One Fathom Bank and off Horsburgh Lighthouse; to introduce associated ships' routeing measures; and to amend the "Rules for Vessels Navigating through the SOMS". The proposal to establish STRAITREP, with Malaysia serving as the VTS authority for STRAITREP Sectors 1 to 6, and Singapore serving as the VTS authority for Sectors 7 to 9, was also adopted. These measures came into force on 1 December 1998, leading to a significant improvement in navigational safety and marine environment protection in the SOMS.

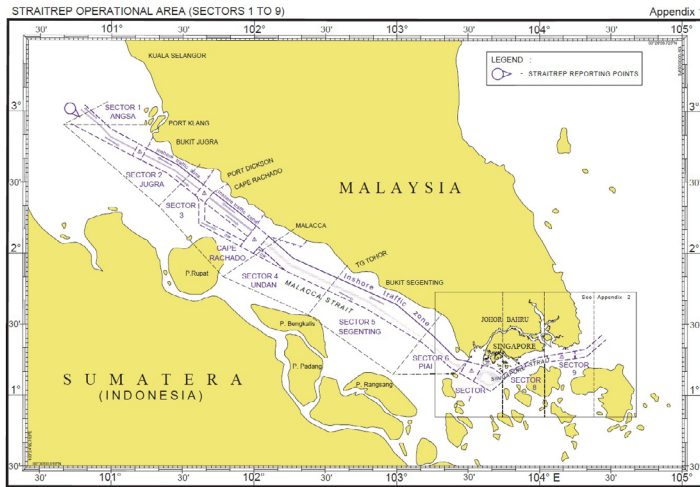


Figure 6.2: Traffic Separation Scheme in the SOMS and STRAITREP Sectors 1 – 9.

Source: Singapore Port Information Year 2026.

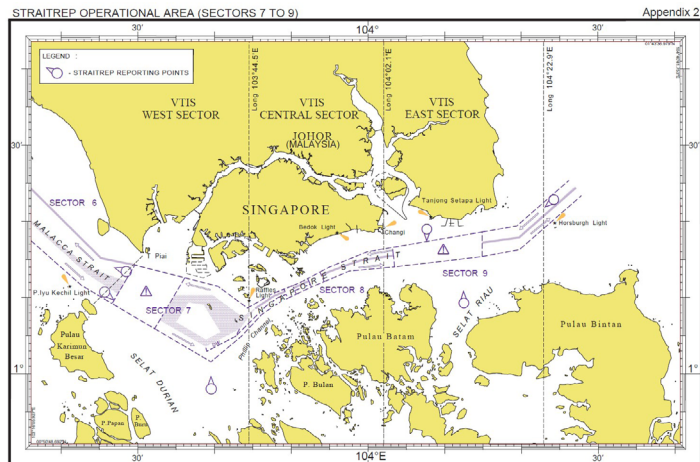


Figure 6.3: Traffic Separation Scheme in the SOMS and STRAITREP Sectors 7 – 9.

Source: Singapore Port Information Year 2026.

Almost 30 years on, this same set of ship routeing and reporting measures remain in place and have stood the test of time – a testament to the robustness of these measures and the good work of the TTEG.

Conclusion

The SOMS has remained open and safe for international maritime passage all these years, thanks to the strong and longstanding technical cooperation among the littoral States, along with the support of users and the IMO. Today, we continue to work together to review existing ships' routeing and ship reporting measures and explore ways to make further enhancements. The spirit of cooperation embodied by the TTEG lives on, more than 50 years since its establishment.

As we chart the next 50 years of tripartite co-operation and beyond, we must continue to uphold the principles and values on which the TTEG was founded. We must build on the work of our predecessors and cannot take our tripartite co-operation for granted. And we must continue to set a positive example for the rest of the world by keeping the SOMS open and safe for all users.

In the words of Prof S. Jayakumar, the then Minister for Foreign Affairs and Minister for Law, at the Institute of Policy Studies/IMO International Conference on the SOMS in 1996: “Whatever we achieve for the Straits of Malacca and Singapore could set a precedent for other Straits of international navigation in the world and thus lead to a safer and less polluted world. Whether the Straits of Malacca and Singapore will continue to serve our children as effectively as it has served us and our forefathers depends on us, the present generation.”

Chapter 7

Protection of the Marine Environment in the Straits of Malacca and Singapore

o/s

Tara Davenport

Introduction

The Straits of Malacca and Singapore (SOMS) is among the most heavily used shipping lanes in the world. The scale of traffic is considerable. Approximately 100,000 vessels passed through the SOMS in 2025, carrying 24 per cent of global seaborne trade and more than US\$2.8 trillion in cargo. A marine environment of considerable ecological and economic value lies beneath this volume of traffic.

The SOMS contains a mosaic of habitats, including coral reefs, mangrove forests, seagrass beds, mudflats, and

sandy beaches. This rich biodiversity serves as breeding grounds for fish, prawns, and crabs, and as habitats for endangered species, including dolphins, dugongs, turtles, and migratory birds. These ecosystems support high biodiversity, providing nursery grounds and feeding areas for numerous fish species, invertebrates, and marine mammals, as well as migratory birds along the East Asian–Australasian Flyway. Mangrove belts and mudflats along the Malaysian and Indonesian coasts of the Malacca Strait host commercially important fisheries and protect coastlines from erosion and storm surges.

The environmental health of the Straits also has strategic significance. For Singapore, a small island state heavily dependent on seaborne trade, maintaining both navigational safety and a clean marine environment is essential to its survival. Malaysia and Indonesia, with long coastlines and extensive coastal populations, similarly view environmental protection as integral to sustainable development and maritime security. Consequently, the marine environment of the SOMS is important not only intrinsically, but also as a foundation for regional stability, economic resilience, and international trade.

This essay explores how Indonesia, Malaysia and Singapore have worked to protect this vital marine ecosystem, cooperating with one another, the International Maritime Organization (IMO), the user States of the Straits and the shipping industry within the framework established by the 1982 UN Convention on the Law of the Sea (UNCLOS).

Environmental Risks from Vessel-Source Pollution

The principal risk to this environment is vessel-source pollution, which arises both from accidents and routine operational discharges. The waters of the Straits are shallow, the seabed is prone to shifting, and the channel is narrow, with the Strait of Singapore measuring barely 3.2 kilometres across at its tightest point. These conditions create navigational hazards for large vessels, and heavy traffic combined with physical constraints increase the risk of ship collisions and groundings. The consequences of a single incident can be severe and transboundary. Several notable incidents have demonstrated the transboundary nature of vessel source pollution in the SOMS. The 7,000 tonnes of crude oil spilled when the *Natuna Sea* grounded in the Singapore

Strait in 2000 settled on Singaporean, Indonesian and Malaysian shores alike; the 2017 *Wan Hai/APL Denver* collision off Johor carried oil into Singapore waters; and the 2021 *Zephyr/Galapagos* collision in the Strait of Malacca, though contained, again straddled the waters of more than one State. Operational pollution, though less conspicuous, is continuous, and includes the discharge of oil, the introduction of invasive species through ballast water, marine debris, and atmospheric emissions.

The Protection of the Marine Environment in Straits under UNCLOS

As explained in previous chapters, the legal regime governing straits used for international navigation is the product of a compromise reached during the Third United Nations Conference on the Law of the Sea (UNCLOS III). The extension of the breadth of the territorial sea from three to 12 nautical miles brought many of the world's most important straits wholly within the territorial seas of the bordering States. The maritime powers were insistent that these waterways remain open to navigation, while the States bordering

straits maintained that the waters were subject to their sovereignty and consequently, the innocent passage regime applied. The compromise embodied in Part III of UNCLOS established the regime of transit passage, a new passage regime that purported to balance the interests of maritime States in freedom of navigation and the security, economic, and environmental concerns of States bordering straits in their territorial sea. Under the transit passage regime, all vessels and aircraft enjoy the right of continuous and expeditious passage, which the bordering State may not deny, hamper or impair and which, unlike innocent passage, they may not suspend. The regulatory authority of States bordering straits over transit passage is not as extensive as its competence over innocent passage in the territorial sea and is limited to certain matters that are expressly set out in UNCLOS.

Three provisions are especially relevant for the protection of the marine environment in straits used for international navigation.

First, Article 42(1)(a) allows States bordering straits to adopt laws on the safety of navigation and the regulation of maritime traffic as specified in UNCLOS Article 41. Article 41 gives States bordering straits the competence

to designate sea lanes and prescribe traffic separation schemes provided they conform to generally accepted international regulations adopted by the IMO. Two or more States proposing sea lanes or traffic separation schemes must cooperate in formulating proposals in consultation with the IMO. The IMO can only adopt sea lanes and traffic separation schemes that are agreed with the States bordering the straits, a reflection of the international importance of such straits for navigation.

Second, Article 42(1)(b) permits States bordering straits to adopt laws and regulations relating to transit passage for the prevention, reduction and control of pollution. This power, however, is qualified in a significant respect: Such laws must give effect to “applicable international regulations” concerning the discharge of oil, oily wastes and other noxious substances, such as those contained in the International Convention for the Prevention of Pollution from Ships (MARPOL 73/78) as amended. These States are not at liberty to impose their own, more stringent pollution standards upon vessels exercising transit passage; they may only apply the standards agreed in international instruments adopted by the IMO.

Third, Article 233 concerns enforcement jurisdiction over vessels, including stopping, boarding, inspecting, and arresting a vessel that has breached the laws of States bordering straits. However, Article 233 permits States bordering straits to take enforcement measures against a vessel on two conditions, both of which must be met. First, only if such a vessel has committed a violation of the laws and regulations referred to in Article 42(1)(a) and (b), i.e., those on safety of navigation and regulation of maritime traffic and those on the prevention, reduction and control of pollution. Second, enforcement jurisdiction can only be exercised where a violation causes or threatens major damage to the marine environment of the strait. The rationale of this limited enforcement jurisdiction is to minimise disruptions to the transit passage of vessels through straits used for international navigation.

Fourth, Article 43 calls upon both the States bordering straits and the user States to cooperate in the establishment and maintenance of navigational aids and in the prevention, reduction, and control of pollution.

Protection of the Marine Environment of the SOMS through the IMO

Singapore, Malaysia and Indonesia, the three littoral States had adopted different positions on the legal status of the SOMS at the beginning of UNCLOS III. Despite this, and as early as 1971, they expressed a common view that the management and responsibility for safety of navigation in the SOMS was the responsibility of the States concerned and required tripartite co-operation. Since then, co-operation, rather than unilateral action, has been the defining feature of the littoral States' approach to the protection of the marine environment in the SOMS. Building upon the foundation laid by the 1971 Joint Statement and the Tripartite Technical Expert Group (TTEG) that was established, and in recognition that navigational safety and environmental protection go hand in hand, the three littoral States have co-operated on a range of navigational-safety and pollution-prevention measures and activities.

Pursuant to UNCLOS Article 41, Indonesia, Malaysia, and Singapore have historically co-operated to submit proposals on traffic separation schemes (TSS) in the SOMS to the IMO. The first TSS was adopted by the

IMO in 1977 and subsequently amended in 1981 and 1998. A minimum under-keel clearance of 3.5 metres was agreed as a compromise between the differing positions of the three States. The TSS organises the flow of shipping into designated lanes that separate vessels travelling in opposite directions and channel deep-draught vessels along the prescribed deep-water route. By imposing order upon what would otherwise be a congested waterway, the scheme reduces the incidence of collisions and groundings that constitute the principal cause of catastrophic oil and chemical spills in the Straits, and keeps the largest vessels clear of the shallow banks and coral reefs that fringe the channel. In practice, the littoral States bordering SOMS have not limited their joint proposals submitted to the IMO to sea lanes and TSS in the SOMS as provided for in Article 41. They have also worked together to submit proposals to the IMO on a series of complementary measures, including the mandatory ship reporting system (STRAITREP), the vessel traffic service, and associated routeing measures. That the number of accidents in the Straits has remained broadly stable notwithstanding the steady increase in traffic suggests that these arrangements have had a measurable effect.

Notably, Malaysia made a proposal for the designation of a Particularly Sensitive Sea Area (PSSAs) to the IMO Marine Environmental Protection Committee (MEPC) in 2015. A PSSA is an area requiring special protection on account of its recognised ecological, socio-economic or scientific significance and its vulnerability to damage from international maritime activities, the protection being effected through Associated Protective Measures (APMs) approved by the IMO. Malaysia's proposal encompassed an area at the southern tip of Peninsular Malaysia in a location that included two coastal protected areas in the State of Johore, Pulau Kukup and Tanjung Piai. The area identified by Malaysia was located within the vicinity of the SOMS TSS and Deepwater Route, in one of the narrowest areas in SOMS, near the entry point to the Port of Tanjung Pelepas and the Straits of Singapore. Malaysia outlined the environmental significance of the area and the need to protect it from international shipping in the SOMS. In April 2017, the Malaysian PSSA Proposal and an Information Paper was submitted to the 71st Session of the IMO MEPC. Malaysia proposed the adoption of two APMs, an area to be avoided and a mandatory no anchoring area. The MEPC did not approve the proposal. Rather, it encouraged Malaysia to continue to consult the other littoral States

and relevant stakeholders at the appropriate forums, in particular on the potential impact of the safety and efficiency of navigation in the TSS of the SOMS, before submitting their proposal. There appears to have been no further action by Malaysia at the IMO concerning the proposal, and it remains to be seen whether this proposal will be revived. While littoral States can make unilateral submissions on PSSAs within their territorial waters, this development serves as an important reminder of the importance of consultation and exchange on such measures located near the SOMS, as reiterated by the MEPC.

Oil Spill Management and the Revolving Fund

The three littoral States have also co-operated with each other on other initiatives to promote the safety of navigation and the protection of the environment. The development of the Marine Electronic Highway enables the real-time exchange of navigational and environmental data and the mapping of oil-spill trajectories. The Revolving Fund Committee (RFC) of the SOMS includes Indonesia, Malaysia and Singapore and the Japan-based non-governmental body, the Malacca Strait Council, and

aims to combat oil spill pollution from ships in the SOMS. Malaysia, Indonesia, or Singapore can draw on the Fund for combating oil pollution caused by ships. The amount drawn will be repaid to the Fund when the State recovers the clean-up costs from the parties responsible. The RFC meets annually to discuss issues relating to oil spills and to update the Standard Operating Procedure for Joint Oil Spill Combat in the Straits of Malacca and Singapore (SOP). The SOP covers areas such as the response areas and division of responsibility among the littoral States, contact points, communication and information sharing procedures, inter-state assistance and reimbursement procedures. The objectives are to facilitate early information sharing and prompt and co-ordinated response to any oil spill incident.

The 2007 Co-operative Mechanism on Safety of Navigation and Environmental Protection

Last but certainly not least, the 2007 SOMS Co-operative Mechanism, adopted pursuant to UNCLOS Article 43, has also been a valuable forum for co-operation and exchange on the protection of the marine environment. The Mechanism comprises three components.

The Co-operation Forum is an annual platform bringing together government officials from user States, shipping industry representatives, and other stakeholders to exchange views with the littoral States and identify priority projects.

The Project Co-ordination Committee (PCC) oversees the implementation of specific projects aimed at promoting safety of navigation and marine environmental protection, to which user States and industry can contribute financially or through technical assistance.

The Aids to Navigation Fund receives voluntary financial contributions from user States, industry, and other stakeholders. The Co-operative Mechanism has enabled states and industry partners to contribute resources and expertise to support initiatives such as upgrading navigational aids, improving hydrographic surveys and charts, enhancing vessel traffic services, and strengthening pollution response capacity. The mechanism is guided by key principles: adherence to international law including UNCLOS; consensus-based decision-making among the three littoral States; openness and inclusivity towards user States; and the recognition of shared interests in keeping the straits safe and environmentally secure. These principles

mark a shift from earlier debates over cost-sharing and responsibility towards a more collaborative, multi-stakeholder approach.

Conclusion

The foremost challenge confronting marine environmental protection in the SOMS is the continued growth of traffic. From approximately 40,000 vessel movements in 1982, transits rose to 94,301 in 2024 and exceeded 100,000 in 2025, with the number of very large crude carriers and ultra-large container vessels increasing steadily. Risks to navigational safety and the marine environment will inevitably increase, a concern compounded by the diversion of vessels from the Red Sea and the Strait of Hormuz. Financing the continuous maintenance and upgrading of navigational aids and environmental protection infrastructure is an ongoing issue, as is ensuring equitable burden-sharing between littoral and user States. Differences in institutional capacity, legal frameworks, and national priorities among the three States may complicate co-ordination. For example, the need for ports in Indonesia and Malaysia with necessary reception facilities to discharge waste has reportedly been an obstacle in the adoption

of Special Areas under MARPOL (International Convention for the Prevention of Pollution from Ships), which would allow special discharge rules to apply to ships passing through the area. Moreover, new threats such as climate change, sea-level rise, and evolving maritime security risks require constant adaptation of environmental protection measures.

Nonetheless, the trajectory of co-operation among Singapore, Malaysia, and Indonesia demonstrates their understanding that the protection of the marine environment of the SOMS is a shared responsibility in the common interests of all States. By leveraging UNCLOS provisions, particularly Articles 41 and 43, and by building practical arrangements such as the TTEG and the Co-operative Mechanism, the three littoral States have created a robust framework for safeguarding this critical maritime corridor. Going forward, continued collaboration among the three littoral States, supported by user States, international organisations, and industry, will be essential to address emerging challenges and ensure that the SOMS remains both a safe highway of global commerce and a resilient marine ecosystem. This balance between use and protection lies at the heart of contemporary ocean governance and underscores the enduring relevance of UNCLOS for the SOMS.

Chapter 8

Managing the Scourge of Maritime Crime in the Straits of Malacca and Singapore: The Story of the Last Two Decades

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Jane Chan

Introduction

“Seaborne robbery cases in the Straits of Malacca and Singapore highest since 2007” was a news headline in early January 2026. This was a reporter’s takeaway after the Regional Cooperation Agreement on Combating Piracy and Armed Robbery against Ships in Asia (ReCAAP) Information Sharing Centre (ISC) released its annual report for 2025.

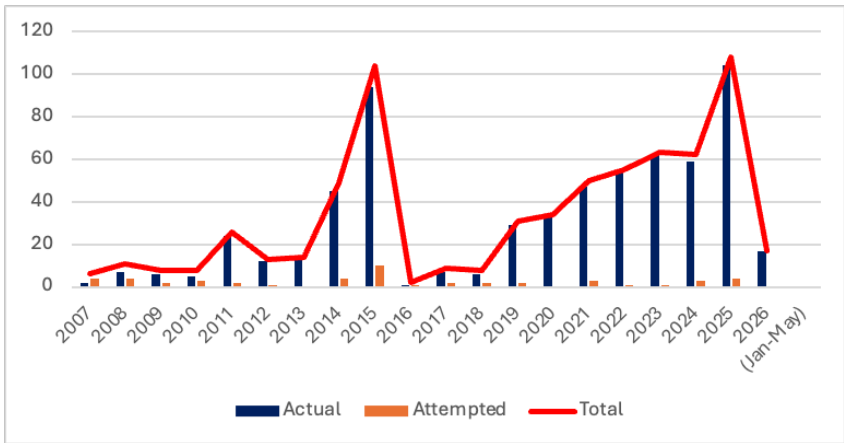


Figure 8.1: Number of reported incidents in SOMS.

Source: ReCAAP ISC.

Aggregated figures can be misleading when assessing the safety and security of this important waterway. Nevertheless, the resurgence of sea robbery incidents in recent years, has highlighted the emergence of an evolving maritime security challenge for the Straits of Malacca and Singapore (SOMS) littorals and the shipping community.

The SOMS are one of the world’s most important maritime arteries, linking the Indian and Pacific Oceans, facilitating approximately half of global total annual seaborne trade tonnage, and serving as a conduit for 70 per cent of Asia’s oil import. Given its strategic and economic significance, maritime security within the SOMS has long attracted international attention.

In the early 2000s, the SOMS were frequently identified as a global piracy hotspot. Attacks ranged from armed robberies and cargo theft to vessel hijackings conducted by organised criminal syndicates. There was heightened concern over the security of the SOMS following the September 11 attacks, with some analysts warning that the SOMS could become a nexus for the potential convergence of piracy and maritime terrorism. In 2005, the London insurance market's Joint War Committee, declared the Malacca Strait to be at risk from "war, strikes, terrorism and related perils". Insurance premiums for transiting through the straits rose sharply. Although the warning was lifted within a year, that episode continues to serve as a reminder on how sensitive and risk averse the market is.

Since then, Indonesia, Malaysia, Singapore and Thailand, which border the SOMS, have achieved considerable success in suppressing organised maritime crimes through various initiatives, including co-ordinated patrols, enhanced maritime domain awareness (MDA) and information sharing, and with new cooperative mechanisms to better engage the shipping industry.

The Evolving Piracy and Sea Robbery Threat

Piracy is defined as occurring on the high seas under international law, whereas incidents that take place within internal waters, territorial waters, and archipelagic waters are generally classified as sea robbery and fall within the jurisdiction of the coastal state. Almost all incidents in SOMS were sea robberies. It is also important to emphasise that the *modus operandi* adopted by the perpetrators in SOMS differs vastly from the pirates off the coast of Somalia/Gulf of Aden, and off the west coast of Africa. The primary objective for those pirates off the coast of Somalia/Gulf of Aden is still kidnapping for ransom, and they are usually armed with automatic weapons and rocket-propelled grenades.

Unlike the organised criminal networks that characterised the trend in early 2000s, incidents in the last decade were largely opportunistic, low-level petty theft from slow-moving vessels, and from vessels lacking ship hardening measures and proper lookout by the crew. Although these incidents rarely disrupt international shipping, the problem continues to be persistent and the criminals highly adaptive.

While the severity of the incidents remained low, the number of reported incidents spiked from 2013 onwards. Perpetrators would board during hours of darkness and mostly target slow moving ships in the vicinity of the Phillip Channel. They usually operate in groups of between one and six men and will generally avoid prolonged confrontation. In some incidents, the perpetrators were reported to be carrying knives or machetes. In most instances, when detected by the crew and the alarm was raised, the perpetrators would escape immediately, often leaving empty-handed. These robbers were mainly after spare parts and personal belongings, which could be easily disposed of on the black market. Most of these incidents reflected opportunistic criminal behaviour rather than organised piracy.

Owing to the efforts by the littorals, by 2016, the number of reported cases had fallen to historically low levels. Unfortunately, the number of reported incidents gradually increased in the years that followed. There were observable differences in the modus operandi of attacks during the last spike in 2015 when compared to recent trends since 2024. While most incidents remained low in severity, there were indications that the perpetrators were getting more sophisticated. There were more

successful high boardings, and more reported use of knives. There were cases where multiple vessels were boarded in proximity over a short period, suggesting a planned operation rather than random opportunistic targets.

The sharp increase in incidents reported since 2024 demonstrates that maritime crime remains resilient despite existing security measures. It is important to emphasise that the severity levels have generally remained low. Most of the crimes were petty thefts and not vessel seizure or cargo hijacking. Most incidents involve bulk carriers and tankers when transiting the Phillip Channel. Nevertheless, the growing volume of attacks creates cumulative risks for maritime safety and security.

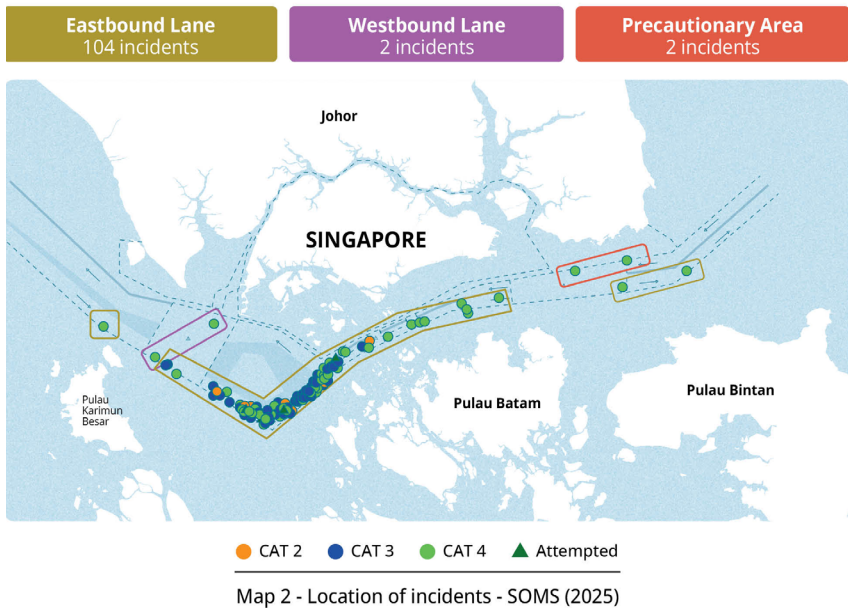


Figure 8.2: Location of incident in SOMS.

Source: ReCAAP ISC 2025 Annual Report.

Littoral State Responses

Faced with international pressure and the threat of foreign naval intervention (the US had proposed the Regional Maritime Security Initiative, which would have entailed US forces patrolling SOMS), Indonesia, Malaysia, and Singapore came together to launch MALSINDO (2004), which later evolved into what is known today as the Malacca Straits Patrol (MSP) since 2008, with Thailand as an added partner.

The MSP framework has three core components: (1) Malacca Straits Sea Patrols, (2) Eyes-in-the-Sky aerial surveillance, and (3) Intelligence Exchange Group. Over the years, MSP has sustained presence and improved deterrence in SOMS. MSP is also widely touted to be a successful example of functional co-operation when there is clear common interest for the partners to come together.

To better manage the problems at hand, Singapore took to enhancing the capabilities of the Police Coast Guard and maritime security agencies. Malaysia established the Malaysian Maritime Enforcement Agency, which started as an offshoot of the navy. Indonesia plans to expand the role of Bakamla and strengthen its naval enforcement activities.

ReCAAP was launched in 2006, and the ISC is based in Singapore. Its core mission as an international organisation is to enhance regional co-operation through information sharing, capacity building, and relevant co-operative arrangements. To date, there are 22 Contracting Parties who have signed on to ReCAAP. Apart from managing a 24/7 reporting centre, ISC also put together threat assessments and trend analysis, to be shared with relevant stakeholders.

This was followed by the establishment of the Information Fusion Centre (IFC) in 2009, led by the Republic of Singapore Navy. A multi-national, multi-agency set up, the IFC's key role is to ensure safe and secure seas through timely and comprehensive information sharing with regional partners. Over the years, IFC has broadened the range of partnerships to include military and enforcement agencies, and shipping companies.

The ISC and the IFC are in the business of maritime domain awareness, and both ensure that the correct information is shared with shippers, relevant navies, and law enforcement agencies that have the jurisdiction to assist.

New Vulnerabilities

While high-end threats have declined, low-level maritime crime has adapted to exploit remaining vulnerabilities. Although most incidents involve petty theft, a sustained increase in the number of incidents could undermine the level of trust and confidence in the littorals' ability to keep the straits safe and secure.

Recent increases in sea robbery incidents suggest that maritime criminality remains an enduring challenge. While contemporary threats differ significantly from those of two decades ago, they nevertheless require sustained policy attention and institutional adaptation. To keep pace with the times, the littorals should invest in new and improved technologies, and deploy advanced surveillance, data fusion, and predictive analytics, as part of their broader efforts to manage the scourge of maritime crime in SOMS.

The shippers are also encouraged to be better prepared when transiting through SOMS. The crew needs to be well trained to ensure good watchkeeping, especially during hours of darkness. Recent arrests, which led to successful prosecution of the perpetrators, were only possible because the vessel was equipped with state-of-the-art surveillance systems. The owner of the ship could produce photographic and video footage as evidence to identify the perpetrators on board the ship.

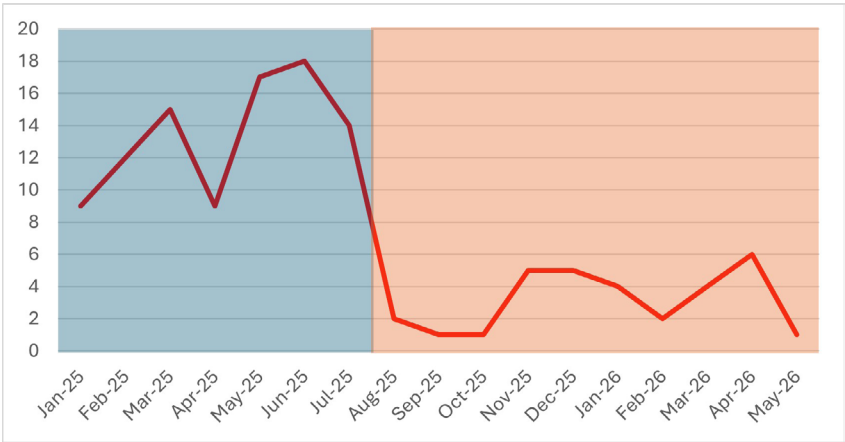


Figure 8.3: Monthly reported incidents in SOMS showing the difference in sea robbery incidents following the arrests of perpetrators in July and August 2025.

Source: ReCAAP ISC.

Conclusion

The security environment in the Straits of Malacca and Singapore has undergone a profound transformation since the early 2000s. Organised maritime crime that once threatened one of the world’s most important maritime corridors has been largely suppressed through effective littoral-state cooperation. This achievement should not be underestimated.

However, the rise of low-level armed robbery and opportunistic maritime crime demonstrates that maritime security remains a dynamic challenge. Existing institutions and operational frameworks remain effective, but there is room for improvement in terms of the level of information sharing and incorporating new MDA tools, to keep up with changing criminal tactics and operational conditions.

The current threat environment requires a transition from a traditional counter-piracy paradigm to a broader maritime crime management approach. It will call for greater trilateral co-ordination, integrated surveillance systems, strengthened legal tools, targeted law-enforcement efforts against shore-based criminal networks, and greater engagement with the maritime industry. The existing littoral-state-led security architecture remains fundamentally sound but requires adaptation to address evolving criminal tactics and operational realities.

Chapter 9

The Co-operative Mechanism for the Straits of Malacca and Singapore

Robert Beckman

Introduction

The rules of international law governing the Straits of Malacca and Singapore (SOMS) and other straits used for international navigation are set out in Part IV of the 1982 United Nations Convention on the Law of the Sea (LOS Convention). Article 43 of the LOS Convention provides that “user States” and “States bordering a strait” should by agreement co-operate on the safety of international navigation in the strait and on the prevention, reduction, and control of pollution in the strait.

The Straits of Malacca and Singapore (SOMS) is the only strait used for international navigation in which a formal mechanism has been established for cooperation as called for in Article 43 of the LOS Convention. The Co-operative Mechanism for the Straits of Malacca and Singapore (Co-operative Mechanism) was formally launched in September 2007 after three years of discussions and negotiations.

In this chapter, I will summarise how the Co-operative Mechanism has operated since its establishment in 2007. I will then trace the history of co-operation between the governments of Indonesia, Malaysia, and Singapore on the Straits of Malacca and Singapore and the negotiations that resulted in the establishment of the Co-operative Mechanism.

The SOMS Co-operative Mechanism

The Co-operative Mechanism comprises three interconnected and complimentary components relating to the safety of navigation and the prevention, reduction, and control of pollution in the straits.

First, a **Co-operation Forum** in which the three littoral States engage in an annual dialogue with user States, the shipping industry, and other stakeholders on issues of common interest on navigation in the SOMS.

Second, a **Project Co-ordination Committee (PCC)** that approves and co-ordinates the implementation of projects to enhance navigational safety and environmental protection in the SOMS. One of the three littoral States serves as the leader or co-leader for each project. User States contribute by either participating in the project or providing financial support. For example, Straits Project 1 on the Removal of Wrecks in the Traffic Separation Scheme was led by Malaysia with the participation of India and Germany. User States that have contributed to and participated in projects include Australia, China, Germany, India, Japan, Republic of Korea, and the United States. In addition, the International Maritime Organization (IMO) and shipping organisations have also participated in the projects.

Third, an **Aids to Navigation Fund**, which receives financial contributions from user States and other stakeholders for the provision and maintenance of aids to navigation in the SOMS. The fund has reportedly

received over US\$20 million in financial contributions, which ensures a sustainable means of financing the maintenance of aids to navigation in the SOMS to facilitate the safety of navigation.

Analysis of the SOMS Co-operative Mechanism

The SOMS Co-operative Mechanism has successfully enhanced the safety of navigation and helped prevent and reduce ship-source pollution in the SOMS. It gives user States and other stakeholders a voice and stake in navigation in the SOMS, but it also recognises that the SOMS is subject to the sovereignty of the three littoral States and the provisions in the LOS Convention. The SOMS Co-Operative Mechanism is consistent with Article 43 of the LOS Convention, even though it includes co-operation between the littoral States and with other stakeholders and user States.

The issue that arises is why the SOMS is the only strait used for international navigation for which a co-operative mechanism has been established, as called for in Article 43 of the LOS Convention. To answer this

question, we must examine the history of co-operation between Indonesia, Malaysia, and Singapore relating to the SOMS from the 1970s to 2007.

History of Co-operation among the Three Littoral States

The Third UN Conference on the Law of the Sea began discussions in 1973 to agree on a new convention governing all uses of the oceans. One of the most important issues that had to be resolved was the breadth of the territorial sea that could be claimed by coastal States. Negotiations leading to the 1958 Geneva Convention on the Territorial Sea and Contiguous Zone could not resolve this issue, so it was silent on the breadth of the territorial sea that could be claimed by coastal States. It was clear from the start of the Third UN Conference that most States would agree that coastal States could claim a territorial sea of 12 nautical miles. One consequence of this would be that many straits used for international navigation, including the Malacca Strait and the Singapore Strait, would fall within the territorial sea of the States bordering the straits.

The governments of Indonesia, Malaysia, and Singapore realised the significance of the negotiations at the Third Conference on straits used for international navigation to passage in the Singapore Strait and the Malacca Strait. Consequently, the three governments entered into discussions on how they could protect their interests under the new legal regime governing straits used for international navigation. The three States agreed that the lower half of Malacca Strait and the Singapore Strait should be treated as a single strait, that responsibility for the safety of navigation in the straits was the responsibility of the three littoral States, and that a formal body for co-operation should be established by the three littoral States.

In the mid-1970s, senior officials of the three States began discussions to address issues relating to navigational safety in SOMS, including under-keel clearance requirements, a traffic separation scheme, and a deep-water route. Recommendations of the senior officials were approved at a ministerial meeting in 1976, and the three littoral States then submitted recommendations to the International Maritime Consultative Organization (IMCO), the UN Specialised Agency on shipping, which was renamed

the International Maritime Organization (IMO) in 1982. The recommendations were accepted, and the IMCO Assembly adopted Resolution A.375 (X) on Navigation through the Straits of Malacca and Singapore on 14 November 1977.

Regular meetings of the technical maritime experts from the three littoral States on the safety of navigation and ship-source pollution have been held since the 1970s, and they are referred to as meetings of the Tripartite Technical Experts Group (TTEG).

Consequently, the three littoral States bordering the Straits of Malacca and Singapore had agreed to co-operate on the safety of navigation and ship-source pollution several years before the LOS Convention was adopted in 1982.

Article 43 of the LOS Convention

Article 43 of the LOS Convention states that user States and States bordering straits used for international navigation “should” by agreement co-operate to enhance the safety of navigation and to prevent, reduce, and control ship-source pollution. Commentators have

suggested that Article 43 was an attempt to encourage user States and the States bordering a strait used for international navigation to share the burden of establishing and maintaining aids to navigation and preventing pollution of the environment from ships passing through the strait. This seems logical and fair, given that user States not only benefit from safe passage through the strait, but their ships also pose a risk of pollution to the marine environment of the States bordering the strait.

Efforts from 1994 to 1999 to Implement Article 43 for the SOMS

After the LOS Convention entered into force in 1994, the three littoral States entered into discussions on the establishment of a co-operative mechanism to implement Article 43 for the SOMS. A series of conferences and workshops were organised by institutions in Singapore and Malaysia from 1994 to 1999. Two of the conferences in Singapore were co-organised by the Institute of Policy Studies in Singapore and the IMO. The Institute of Policy Studies was then headed by Ambassador-at-Large, Tommy Koh, the former President of the Third UN Conference on the Law of the Sea.

At the 1999 Conference, the nature and extent of the assistance that littoral States should seek from user States on the implementation of Article 43 was discussed. However, no meeting of senior officials from the three littoral States was held, and the issue of how to implement article 43 was indefinitely deferred because of three developments: (1) the Asian financial crisis from 1997 to 2002; (2) the political upheavals in Indonesia after the fall of President Suharto in 1998; and (3) the 2003 Asian SARS crisis.

Events Triggering a Renewed Discussions after 2001

Three events in the early 2000s triggered increased concern about international shipping in SOMS and resulted in renewed discussions on establishing a co-operation mechanism for the SOMS.

First, the terrorist attack on the World Trade Center in New York on 11 September 2001, resulted in increased concern in the shipping community about the threat of “maritime terrorism”, and “maritime security” suddenly became a priority of the IMO. Concerns were raised

about the prospect of a terrorist attack closing major international straits such as SOMS.

Second, the number of incidents of attacks on ships in the Straits of Malacca and Singapore increased dramatically after the Asian Financial Crisis, and Lloyd's declared the Malacca Strait a "war risk" zone for the purposes of marine insurance.

Third, in 2004, Japan made it known through informal channels that it could no longer bear the costs of maintaining and replacing the aids to navigation in SOMS, and that assistance from other States would be necessary. Japan had made substantial contributions in the past through the Malacca Strait Council of Japan to maintain and replace the aids to navigation in the SOMS.

As part of its effort to address the threat of maritime terrorism, the IMO Council decided that the IMO could play a vital role in protecting sea lanes of strategic importance. The IMO Secretariat then undertook a study of vital shipping lanes and gave the SOMS special consideration.

In March 2005, the Secretary-General of the IMO delivered the keynote address in Singapore at the ASEAN

Regional Forum on Confidence-Building Measures and stated that the IMO had initiated a round of consultations with the SOMS three littoral States and selected user States. He advised that an IMO-sponsored meeting would be held in Jakarta in September 2005 and opined that the September meeting would provide a unique opportunity for the littoral States, user States, and other stakeholders to enhance navigational safety, environmental protection, and security in the SOMS.

The Batam Joint Ministerial Statement, August 2005

In August 2005, one month before the IMO-organised meeting in Jakarta, the foreign ministers of Indonesia, Malaysia, and Singapore met in Batam to discuss matters relating to safety, security, and environmental protection in the SOMS. This was the first meeting of the three foreign ministers on SOMS since 1977.

Subsequently, the three foreign ministers issued the Batam Joint Ministerial Statement of August 2005, which is commonly referred to as the 2005 Batam Statement. The 2005 Batam Statement set out the agreed principles of co-operation in the SOMS and stated: (1) that the

littoral States have sovereignty and sovereign rights in the SOMS; (2) that the littoral States have primary responsibility over safety, environmental protection, and security in the SOMS; and (3) that any measures adopted or taken on these matters must be in accordance with international law, including the LOS Convention. Thus, at the meeting in Batam the three States appeared to have agreed on the principles under which they would work with user States and the IMO to implement co-operation pursuant to Article 43 of the LOS Convention.

IMO Meetings in Jakarta in 2005, Kuala Lumpur, in 2006 and Singapore in 2007

Meetings were convened by the IMO in Jakarta in 2005, Kuala Lumpur in 2006, and Singapore in 2007. The meetings were attended by the three littoral States and interested user States. The net result of the three meetings was the establishment of a SOMS Co-operative Mechanism under the direction and control of the three littoral States.

It should be noted that although the impetus for the establishment of the Co-operative Mechanism came

from concerns of the IMO on the security of shipping in the SOMS, the Co-operative Mechanism does not include co-operation on security. It is limited to co-operation on navigational safety and protection of the marine environment. This limitation is justified for at least two reasons. First, Article 43 calls only for cooperation on safety and pollution, not security. Second, maritime security issues involve completely different agencies, including the ministries of defence and foreign affairs, whereas the safety of navigation and ship-source pollution are the responsibility of the marine department of the Ministry of Transport.

Conclusion

The Co-operative Mechanism in the SOMS is a model for how States bordering straits used for international navigation can co-operate with each other and with user States in accordance with the LOS Convention to enhance navigational safety and protect the marine environment while safeguarding their territorial sovereignty.



Figure 9.1: Participants at the 16th Co-operative Forum Meeting in Penang, Malaysia (29 to 30 September 2025). (L to R) Capt Abdul Samad Shaik Osman, Deputy Director General of Marine (Operations), Malaysia Marine Department (MMD); Mr Hernadi Tri Cahyanto, Director of Navigation, Directorate General of Sea Transportation, Indonesia; Datuk Haji Dickson Dollah, Director General of Marine, MMD; Mr Ang Wee Keong, Chief Executive, Maritime and Port Authority of Singapore; and Capt Nazri Abu Hassan, Deputy Director General of Marine (Planning), MMD.

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